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The aspartate-to-homoserine pathway recurs across maize grain amino acids, soil nitrogen, and yield selection

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Abstract

Grain nitrogen in maize depends on nitrogen uptake, assimilation, and allocation, but it is unclear whether the same metabolic functions underlie natural variation in grain amino acid composition, genetic variation associated with soil nitrogen, and response to selection. We tested this in three populations, each with a distinct nitrogen-related phenotype. In the Goodman–Buckler Association Panel, grain free amino acids were dominated by asparagine and proline and were highly heritable, and genome-wide association mapping of 71 traits implicated genes for amino acid metabolism and transport. The loci shared across the most traits lay near a peptidase A1 gene and endoplasmic reticulum membrane protein genes. In 3,107 georeferenced landraces from across the Americas, association mapping of soil nitrogen at collection sites identified variants near an aspartate kinase isozyme and an omega-amidase that passed tests for confounding with geography and climate. Indian Chief and Jarvis responded to 14 cycles of reciprocal recurrent selection for yield with different changes in nitrogen allocation. The two populations diverged genetically, and Indian Chief accumulated more free amino acids, mostly proline, while protein-bound amino acids were unchanged. Their candidate regions included protein-turnover genes in Indian Chief and amino acid importers in Jarvis, consistent with these population-specific phenotypes. Each of the four analyses highlighted a different enzyme of the aspartate-to-homoserine pathway, the only pathway that remained significant after correction across all pathways. Together, these results point to aspartate-family metabolism as a shared source of candidate genes for how nitrogen is partitioned among grain amino acid pools in maize.

Keywords maize, grain nitrogen, free amino acids, aspartate-family amino acids, genome-wide association, reciprocal recurrent selection

Introduction

Nitrogen (N) is essential for maize growth and kernel development, but its use in agriculture is a major sustainability challenge. Crop production relies heavily on N fertilizer, yet much of the applied N is not recovered by the crop and is lost through leaching, runoff, or gaseous emissions (Ojeda-Rivera et al., 2025). Reducing these losses requires understanding how plants acquire, assimilate, transport, and remobilize N, and how they allocate it to the developing grain (Masclaux-Daubresse et al., 2010a; Havé et al., 2017). In maize kernels, N is held in free amino acids (FAAs) and protein-bound amino acids (PBAAs), so grain amino acid composition is a direct biochemical readout of how N is allocated to the grain.

Plant N metabolism centers on a few key amino acids. Ammonium is assimilated via the GS/GOGAT cycle, producing glutamine and regenerating glutamate from 2-oxoglutarate, so Glu/Gln metabolism links N assimilation with carbon metabolism (Masclaux-Daubresse et al., 2010a; Hodges, 2002). Glutamine and asparagine are the main N-rich transport and storage forms, while glutamate and aspartate donate amino groups to carbon skeletons (Bernard and Habash, 2009; Masclaux-Daubresse et al., 2010a). Aspartate is particularly important because it feeds asparagine and the aspartate-family amino acids (methionine, threonine, lysine, isoleucine) (Viola, 2001; Wang et al., 2018; Han et al., 2021). During grain filling, protein degradation in senescing leaves releases amino acids that supply grain N (Havé et al., 2017; Liu et al., 2008). Proline adds a stress-related component, with genotype-dependent accumulation and export from maize leaves under water stress (Carceller et al., 1999). Thus, shifts in grain FAA composition may reflect redistribution among multiple N pools, not just changes in total N.

Most grain N is stored in storage proteins, so protein-bound amino acid composition is largely determined by the proteins produced. The free amino acid pool is small but reflects amino acids delivered to

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and metabolized in the developing kernel, and may therefore better indicate N supply. How much this pool varies heritably among lines, and which genes control that variation, is less well known than for protein-bound amino acids (Shrestha et al., 2022).

Maize germplasm enables grain N metabolism to be studied in three ways. First, standing variation among inbred lines. The Goodman–Buckler panel captures much of the allelic diversity in public breeding and has been used to map grain composition traits (Flint-Garcia et al., 2005; Shrestha et al., 2022). Second, adaptation to soil. The landraces evolved under local conditions across the Americas, so alleles affecting N acquisition or use may correlate with soil N at collection sites. Georeferenced landraces enable environmental GWAS (Romero Navarro et al., 2017), but soil N covaries with geography and climate and must be controlled. Third, selection for yield. Fourteen cycles of reciprocal recurrent selection in Indian Chief and Jarvis altered N uptake and allocation (Comstock et al., 1949; Moll and Stuber, 1971). Jarvis increased grain N via post-anthesis uptake, while Indian Chief retained more N in stover (Moll et al., 1994). Comparing Cycle 0 vs. Cycle 14 highlights regions with large allele-frequency shifts, though drift is also substantial in small populations (Gerke et al., 2015).

These populations differ in germplasm, in the N-related phenotype measured, and in how genetic signals are detected, so the same genes are not expected to appear in all of them. Each analysis also yields hundreds to thousands of candidate genes, many of which overlap by chance. If N metabolism shapes all three, however, different genes in the same metabolic pathways could be implicated. We therefore compared the analyses at the pathway level, asking whether any pathway contained candidates from more analyses than expected by chance.

We combined grain amino acid profiling and GWAS in the Goodman–Buckler panel, soil-N environmental GWAS in landraces, genomic population scans of Indian Chief and Jarvis, and reanalyzed the B64 × CML14 biparental population developed for grain asparagine variation for aspartate and proline. We analyzed free and protein-bound amino acid pools, focusing on asparagine, aspartate, glutamate, glutamine, and proline because of their roles in N transport, C–N metabolism, and storage. Although candidate regions overlapped no more than expected by chance, each analysis implicated a different enzyme of the aspartate-to-homoserine pathway, the only pathway that remained significant after correction. Beyond this pathway, individual candidates pointed to the aspartate–glutamate junction, where amino acid N connects with C–N recycling, and to amino acid transport and protein turnover, with different functions highlighted in each population. Together, these results suggest that aspartate-family metabolism is a shared source of variation in maize N metabolism.

Results

Kernel free amino acid composition in the Goodman–Buckler panel is dominated by asparagine and proline

Amino acids are the main form in which nitrogen is transported and stored during remobilization (Masclaux-Daubresse et al., 2010b; Santiago and Tegeder, 2020). During senescence and grain filling, proteins in source leaves are degraded by autophagy and vacuolar proteases (Potter and Feller, 2026), and the released nitrogen is moved to the grain largely as the amides Gln and Asn (Teixeira et al., 2026), loaded into the phloem by transporters such as AAPs and UmamiTs (Santiago and Tegeder, 2020). Most of the seed nitrogen can come from these remobilized sources (Masclaux-Daubresse et al., 2010b). The free amino acid pool of the grain is therefore the soluble, metabolically active fraction of grain nitrogen through which delivered nitrogen passes before it is stored in protein, and its composition can reveal variation in how nitrogen is supplied and used that total protein alone does not show.

FAA composition was evaluated across 280 of the 282 lines in the Goodman–Buckler panel grown in four replicates (two years × two blocks) at the Genetics Farm near Columbia, Missouri (Supplemental Tables S1–S2). To assess relative abundance, amino acids were ranked within each genotype based on average values across replicates. Asn and Pro consistently showed the highest relative abundance (Supplemental Figure S1A–B). Asn ranked as the most abundant amino acid in 146 genotypes (52%), while proline ranked first in 118 genotypes (42%). Notably, Asn and Pro comprised the top two amino acids in 179 of 280 genotypes (64%), regardless of order (Supplemental Figure S1B). This rank dominance reflected large differences in abundance. Asn was about 144 times as abundant as Cys, with Pro next, and Met and Cys at the low end (Supplemental Figure S1C). How abundant an amino acid was did not predict how much it varied among lines (Supplemental Table S3). Met varied the most (coefficient of variation, CV, 180%) despite its low abundance, followed by Gln (CV 130%). In contrast, Glu, Trp, and Gly varied the least (CV 44–54%). Asn and Pro were moderately variable (CV 70% and 63%, close to the median of 72% across the 20 FAAs), but their absolute differences among lines were still large.

Differences among lines were consistent across years. For the 251 lines grown in both years, values were correlated between years for all free amino acids (Spearman ρ = 0.33–0.75, median 0.56; e.g., Asn 0.65, Pro 0.67; Supplemental Table S3). Median broad-sense heritability (H^2) was 0.75 for the 20 FAAs (0.47–0.88), compared to 0.60 (0.49–0.68) for the 15 PBAAs measured by Shrestha et al. (2022), estimated with the same model, indicating that most variation among line means reflects genotype rather than field noise. The PBAA set is smaller because acid hydrolysis converts Asn to Asp and Gln to Glu (reported together as Asx and Glx) and destroys Trp and Cys, and Gly was not reported. The most heritable FAAs were Lys (0.88), Asn (0.83), Asp (0.81), and Pro (0.80). The most variable (Met, Gln) were also measured reliably (0.78, 0.69). Median genomic heritability (h^2) from the marker-based relationship matrix was 0.48 for FAAs and 0.34 for PBAAs, suggesting substantial additive variation suitable for association mapping (Supplemental Figure S2; Supplemental Table S4, including ratio traits).

To quantify the free pool's contribution to grain amino acid N, we compared genotype level FAAs and PBAAs in matched taxa (n = 279; Supplemental Figure S3, Supplemental Table S5) (Shrestha et al., 2022). The protein-bound pool dominated, exceeding the free pool 47-fold for total amino acids and 42-fold for amino acid-derived N (each amino acid weighted by its number of N atoms; paired Wilcoxon tests, all $P < 10^{-40}$), so FAAs contributed only 2.4% of grain amino acid-derived N. Pool composition, however, differed. To compare the two pools on the same basis, we combined free Asp+Asn and Glu+Gln to match Asx and Glx and excluded Gly, Trp, and Cys (8.4% of the free pool) from FAA totals. On this basis, Pro comprised 23.1% of FAAs versus 10.6% of PBAAs, and Asx+Glx+Arg+Lys+His comprised 56.0% versus 26.5%. Considering only the N-rich residues that are

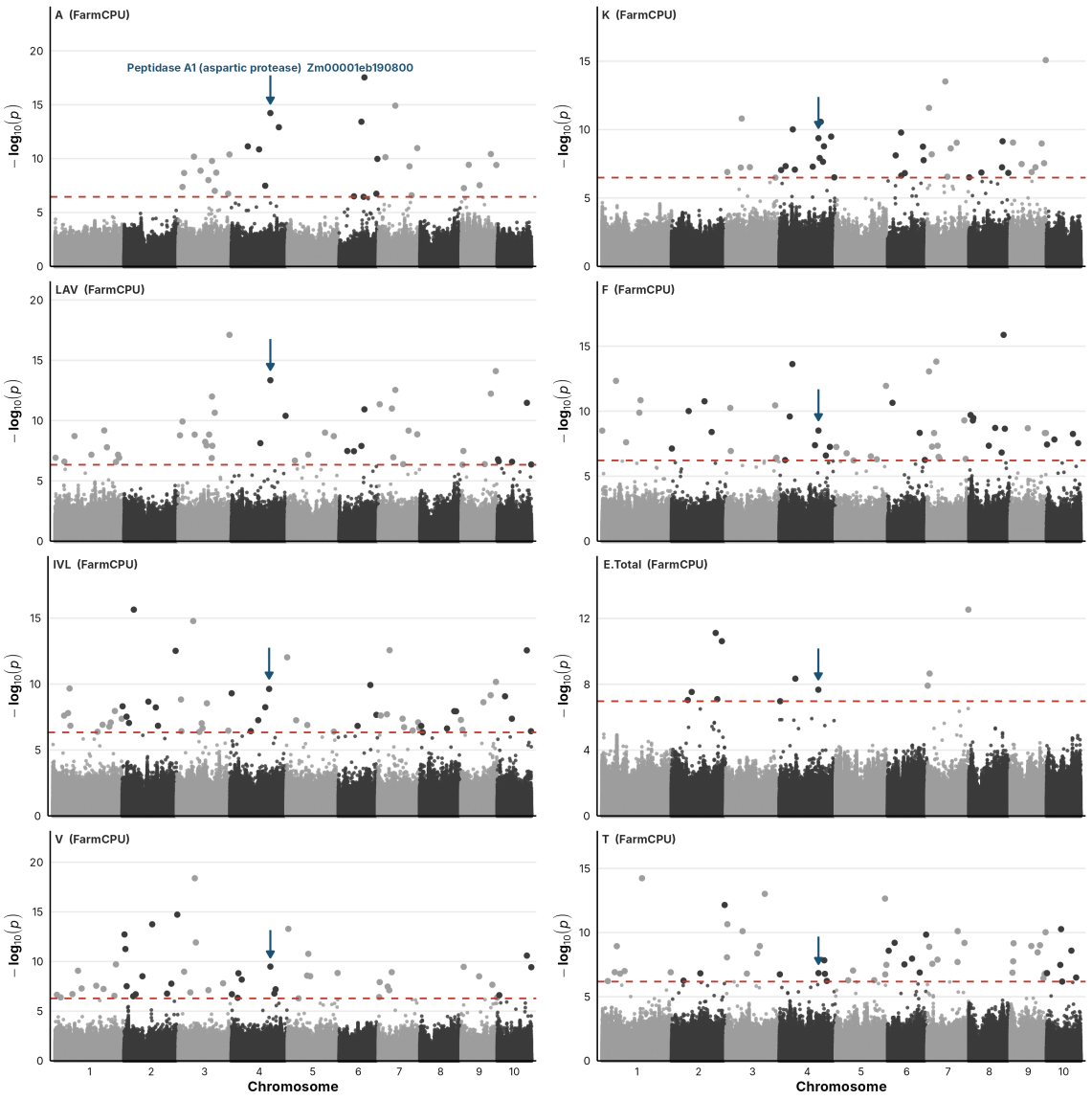


Figure 1 Manhattan plots associated with eight amino acid traits link to an aspartic protease gene A chromosome 4 locus associated with alanine (*A*), *LAV*, *IVL*, valine (*V*), lysine (*K*), phenylalanine (*F*), *E.Total*, and threonine (*T*) lies adjacent to a peptidase A1 (aspartic protease) gene. FarmCPU Manhattan plots show the significant traits at chr4:174.55–174.57 Mb. Arrows indicate the top SNP within *Zm00001eb190800*. Red dashed lines mark the 5% Benjamini–Hochberg FDR thresholds.

resolved in the free pool, Asn+Gln+Arg+Lys+His accounted for 35.7% of FAAs. Genotypic variation also differed. The Pro fraction had a coefficient of variation (CV) of 44% among lines in FAAs compared to 8% in PBAAs, and the amide+basic fraction was 20% compared to 6% (Supplemental Table S5). Thus, the free pool is small but compositionally variable among lines and consistent across years, dominated by Asn and Pro, and enriched for N-rich residues, whereas the protein-bound composition varies much less. Because the differences in free amino acids among lines were large and reproducible across years, the free pool was well suited for association mapping.

Association mapping identifies amino acid metabolism and transport candidates for abundant, heritable free amino acids

To identify loci affecting grain amino acid composition, we defined 78 amino acid–derived phenotypes (Supplemental Tables S1–S2), including the absolute abundance of 20 individual FAAs; six biosynthetic-family composites (IMNTDK, EHPRQ, SGC, WFY, LAV, IVL); 20 ratios to the total free pool; 25 ratios to the family composite; four ratios between amino acids (N.D, N.E, N.Q, Q.E); and three whole-pool measures (total FAA, total PBAA (Shrestha et al., 2022), and estimated nitrogen).

We ran GWAS using three models (MLM, BLINK, and FarmCPU). At a BH FDR of 5%, we found 2,461 significant SNPs across 71 phenotypes, forming 2,152 loci and yielding 3,693 candidate genes (Supplemental Tables S6–S7). Most candidate genes were specific to

one phenotype (3,002), 691 were shared by two or more phenotypes, and only 193 by more than two. The number of candidate genes per phenotype was similar across biosynthetic families (71, 69, and 67 for Asp-, Glu-, and Ser-family phenotypes), so families with more phenotypes simply had more candidate genes. The loci associated with the most phenotypes were not near amino acid biosynthesis genes. Three loci were each associated with nine phenotypes. The locus on chromosome 4 (212.55 Mb) had no gene within 25 kb, and the locus on chromosome 7 (40.80 Mb) was near a K homology domain protein gene (*Zm00001eb306240*). The locus on chromosome 9 (133.93 Mb) contained four genes, including an ER membrane protein complex subunit 4 gene (*Zm00001eb394360*) and a rhomboid-like protein gene (*Zm00001eb394380*), but its nine phenotypes were explained by only three groups of SNPs (Supplemental Figure S4), because many phenotypes are ratios or composites of the same amino acids. A fourth locus on chromosome 4 at 174.57 Mb was associated with eight phenotypes from four biosynthetic families (*A*, *V*, *K*, *F*, *T*, *LAV*, *IVL*, and *E.Total*), with significant SNPs 2.6–6.1 kb upstream and 5.9 kb downstream of a peptidase A1 (aspartic protease) gene (*Zm00001eb190800*; Figure 1).

Furthermore, we investigated individual FAAs (Table 1). Asn, the most abundant ($H^2 = 0.83$, $h_g^2 = 0.68$), had the most loci (356 across abundance and five ratios) and candidates, including two asparagine synthetase domain genes (*Zm00001eb036060*, *Zm00001eb036070*; *N.E*), a glutamine amidotransferase type 2 (*Zm00001eb329710*; *N.E*, $P = 6.1 \times 10^{-10}$), homoserine dehydrogenase (*Zm00001eb405410*; *N.IMNTDK*, SNP within gene), and lysine-branch enzymes dihydrodipicolinate synthase/reductase (*Zm00001eb389290*, *Zm00001eb176170*). An AVT6C transporter (*Zm00001eb368680*) was significant for Asn ($P = 1.3 \times 10^{-8}$) and the Asp-family composite ($P = 2.5 \times 10^{-11}$) in both BLINK and FarmCPU. For lysine, the most heritable FAA ($H^2 = 0.88$, $h_g^2 = 0.73$), a SNP for lysine as a fraction of the total free pool (*K.Total*; $P = 2.5 \times 10^{-11}$; significant in BLINK and FarmCPU) lies in *Zm00001eb277160*, an NRT1/PTR family 4.6 transporter gene. Another SNP for lysine relative to aspartate-family amino acids (*K.IMNTDK*; $P = 2.8 \times 10^{-11}$) is 15 kb from *Zm00001eb360480*, which encodes NADH-dependent glutamate synthase.

Pro, the second most abundant FAA, was nearly as heritable as Asn ($H^2 = 0.80$), but it had a lower genomic heritability ($h_g^2 = 0.40$), suggesting that the markers capture less of its variation among lines. It also had fewer loci (59). However, its ratio traits had no significant associations. One proline SNP lies within a class I glutamine amidotransferase gene (*Zm00001eb217220*; $P = 2.2 \times 10^{-9}$). Glutamine, the second most variable FAA (CV 130%) with high genomic heritability (0.85), showed the strongest association among the amino acids in Table 1 ($P = 4.9 \times 10^{-22}$, *Q.E*). A SNP for free glutamine (*Q*; $P = 1.2 \times 10^{-11}$) lies within a WAT1-related (UmamiT) gene on chromosome 9 (*Zm00001eb379220*). A SNP for the Gln/Glu ratio (*Q.E*; $P = 8.4 \times 10^{-8}$) lies within the amino acid/auxin permease gene *aaap40* (*Zm00001eb261470*) on chromosome 6, in a 240-kb region that also contains the lysine histidine transporters *LHT1* (*Zm00001eb261480*) and *LHT2* (*Zm00001eb261440*) and three further amino acid permeases (*aaap41*, *aaap43*, *aaap44*; *Zm00001eb261490–Zm00001eb261510*). Methionine, the most variable FAA (CV 180%), had 104 loci but no trait-matched candidate gene. Its high CV likely reflects a strongly skewed distribution across lines, where the highest line had about 200 times as much Met as the lowest (Supplemental Table S3). Thus, genes involved in Asn/Asp-family metabolism and amino acid transport linked to their respective amino acids, whereas loci common to multiple phenotypes suggested roles in protein turnover rather than in single biosynthetic reactions.

To test whether the free Asp and Pro variation seen in the panel also segregates in a biparental cross, we mapped both traits in a B64 × CML14 $F_{2:3}$ population developed to map kernel free Asn, in which Asn is controlled mainly by a single QTL at *ASN3* on chromosome 1 (Fitzsimmons et al., 2026). Because B64 and CML14 were chosen for Asn rather than Asp or Pro, this analysis was exploratory. In the Goodman–Buckler panel, CML14 was among the lowest lines for free Asp, and B64 was near the median, with both being below the median for Pro (Supplemental Figure S5A,C). We identified a single QTL for each trait at the same position on chromosome 3 (peak at 35.99 Mb), with LOD 6.12 for Asp (threshold 3.70, 1,000 permutations) and 4.55 for Pro (threshold 3.63), explaining 10% and 7% of phenotypic variance, respectively (Supplemental Figure S5B,D). This shared QTL was distinct from the Asn QTL, indicating that free Asp and Pro share genetic control that is separable from Asn in this cross. The 1.5-LOD support intervals for this shared peak were similar in size and location, spanning 14.25 and 15.27 cM for Pro and Asp, respectively, and contained 1,550 and 1,578 genes (Supplemental Table S8). The interval spans the low-recombination region around the chromosome 3 centromere, so it cannot resolve a causal gene. However, within 1.5 cM of the peak, it includes the tandem pair of CC-type glutaredoxin genes (*Zm00001eb131830*, *Zm00001eb131840*) and the LHT-like amino acid transporter (*Zm00001eb132750*), all of which also lie in highly differentiated Jarvis windows (see below). Of these, only the LHT-like gene is consistently expressed in developing kernels (Supplemental Table S8).

Soil nitrogen environment GWAS of maize landraces points to candidate genes in aspartate-family and amide nitrogen metabolism

In the Goodman–Buckler panel, the free amino acid pool of the grain was dominated by Asn and Pro, which, along with Lys and Asp, were also among the most heritable free amino acids. Association mapping identified 3,693 candidate genes, including genes involved in Asn and Asp-family metabolism and amino acid transport. Because amino acids are the main transport form of nitrogen in the plant, their metabolism may also respond to nitrogen availability in the soil. However, these inbred lines represent standing variation in temperate germplasm grown under managed fertility. We therefore turned to maize landraces collected across a soil nitrogen gradient (Figure 2B) and asked whether genetic variation associated with soil N implicates similar nitrogen-metabolism functions. Because soil N reflects the historical collection environment rather than a grain phenotype, we did not equate soil N adaptation with grain amino acid composition.

We associated the georeferenced maize landrace panel (Romero Navarro et al., 2017) with soil N at each collection site (Poggio et al., 2021), using 0–5 cm as the primary phenotype and 5–15 cm for validation (highly correlated; $r = 0.84$ across 3,107 accessions; Figure 2A–B). Soil N is a property of the collection site, so accessions from the same or nearby sites share both soil N values (467 distinct values among 3,107 accessions) and ancestry. As a result, genomic heritability was ∼1.00 and remained 0.98 after removing 50 genotype PCs (permuted $h^2 \leq 0.05$; Figure 2C), reflecting shared ancestry and location

Table 1 Association mapping of the focal FAAs in the Goodman–Buckler panel. Median, coefficient of variation (CV), line-mean broad-sense heritability (H^2) and genomic heritability (h_g^2) are from Supplemental Table S3. Loci: loci (Supplemental Table S7) significant at 5% Benjamini–Hochberg FDR for at least one phenotype reporting on the amino acid; in parentheses, loci for the absolute trait alone. Best P: smallest P among those phenotypes. Trait-matched candidate genes: genes within 25 kb of a significant SNP with their putative function.

FAA	Median	CV (%)	H^2	h_g^2	Loci	Best P	Trait-matched candidate genes (putative function)
Asparagine (N)	3.783	70	0.83	0.68	356 (73)	2.8×10^{-20}	*Zm00001eb368680* (amino acid transporter AVT6C); *Zm00001eb329710* (glutamine amidotransferase type 2); *Zm00001eb405410* (homoserine dehydrogenase); *Zm00001eb036060* (asparagine synthetase domain); *Zm00001eb036070* (asparagine synthetase domain); *Zm00001eb389290* (dihydrodipicolinate synthase); *Zm00001eb176170* (dihydrodipicolinate reductase)
Proline (P)	3.202	63	0.80	0.40	59 (59)	2.7×10^{-15}	*Zm00001eb217220* (glutamine amidotransferase)
Lysine (K)	0.272	90	0.88	0.73	116 (54)	8.3×10^{-16}	*Zm00001eb277160* (NRT1/PTR family 4.6); *Zm00001eb360480* (glutamate synthase (NADH)); *Zm00001eb352400* (WAT1-related (UmamiT))
Aspartate (D)	1.178	76	0.81	0.84	91 (20)	3.3×10^{-16}	*Zm00001eb207630* (WAT1-related (UmamiT))
Glutamine (Q)	0.371	130	0.69	0.85	165 (65)	4.9×10^{-22}	*Zm00001eb379220* (WAT1-related (UmamiT)); *Zm00001eb261470* (amino acid transporter); *Zm00001eb261480* (amino acid transporter); *Zm00001eb261490* (amino acid transporter)
Methionine (M)	0.047	180	0.78	0.46	104 (56)	3.2×10^{-17}	none

rather than heritable variation in the plants. We therefore tested for confounding by relatedness and geography directly. Similarly, across three GWAS (MLM, BLINK, FarmCPU) models, 234 SNPs were significant at 5% FDR (Figure 2D). BLINK detected 218, FarmCPU 30, and MLM 2. Most SNPs were detected by a single model (203 by BLINK and 16 by FarmCPU), and 15 were detected by at least two, 13 by BLINK and FarmCPU, one by MLM and BLINK, and one (S5_22338979) by all three. In total, 448 genes lie within 25 kb of 216 of these SNPs (Supplemental Table S9).

Because soil N varies across the landscape and landrace ancestry follows geography, an association with soil N can arise from shared ancestry or from other site variables that covary with soil N, rather than from soil N itself. We therefore filtered the soil N SNPs in several steps, using two tests to assess whether associations reflected soil N rather than geography or climate, and then focusing on SNPs detected by more than one model. The first asked whether the same SNPs were detected when other site variables were used as the GWAS phenotype. We ran GWAS with the same models for latitude, longitude, elevation, soil pH, CEC, and aridity and counted SNPs shared with soil N (Supplemental Figure S6; Supplemental Tables S10–S11). Only 2 of the 234 soil-N SNPs were also significant for latitude, longitude, or elevation, close to the 0.7 expected by chance ($P = 0.14$). By contrast, soil N shared many SNPs with aridity (38 compared to 0.2 expected) and soil pH (13 compared to 0.1), which covary with it. In total, 190 of the 234 SNPs were not significant for any other variable. Because a SNP can follow a geographic gradient without reaching significance in another GWAS, the second test did not use a GWAS model. For every SNP in the genome, we calculated the correlation between genotype and each site variable across the 3,107 accessions and considered a soil N SNP linked to another variable if its correlation was in the top 1% of the genome for that variable (Supplemental Figure S7; Supplemental Tables S12–S13). By this measure, 13.2% of soil N GWAS SNPs were linked to a geographic variable, compared with 1.8% of all SNPs and 35.9% of SNPs selected only for their correlation with soil N ($P = 5.9 \times 10^{-15}$). The GWAS models therefore removed much, but not all, of the geographic signal. Overall, 168 of the 234 SNPs passed both tests. Because associations detected by a single model are less reliable, we focused on the 15 SNPs detected by at least two models. Of these, 14 passed the first test and 12 passed both (Table 2). The three that failed included the strongest signal (S9_151742636; $P = 2.6 \times 10^{-13}$), which was also significant for aridity and soil pH. Of the 12, five had genotypes correlated with soil N but not with other site variables. The other seven had genotypes only weakly correlated with soil N itself, so the lack of correlation with other variables was uninformative for them.

Four soil N SNPs map near genes involved in Asp-family or amide nitrogen metabolism. S5_22338979 lies 4.4 kb upstream of *Zm00001eb220040*, an amino acid kinase-domain gene annotated with aspartate kinase activity, the first step of the aspartate-derived amino acid pathway. It is the only soil-N SNP significant in all three models and is also significant at 5–15 cm. S1_16128632 lies within *Zm00001eb005710*, an ω-amidase/NIT2-like CN-hydrolase (Zhang and Marsolais, 2014). Both SNPs are rare (minor allele frequencies of 0.030 and 0.016), so their associations rest on few accessions. Because both SNPs are rare, we examined where the minor-allele carriers were collected (Supplemental Figure S8). Carriers of the ω-amidase allele came from 56 collection sites, and carriers of the aspartate kinase allele came from 115 sites, each spanning 11 countries, so neither association rests on a single site or local population. Both alleles were concentrated in Mexico and Central America (87% and 93% of carriers, compared with 74% of the panel) and were nearly absent from South America (5 of 667 accessions for each allele). Carriers came from soils with more N than non-carriers (median 414 compared with 251 for ω-amidase and 356 compared with 248 for aspartate kinase), and both heterozygous and homozygous carriers showed this shift (Kruskal–Wallis $P = 4.4 \times 10^{-9}$ and 3.3×10^{-10}). Because soil N also differs between regions, we compared carriers with non-carriers within regions. Within Mexico, ω-amidase carriers came from soils with nearly twice the N of non-carriers (median 424 compared with 233, 28 carriers, Wilcoxon $P = 1.0 \times 10^{-4}$), whereas the difference within Central America was smaller (451 compared with 404, 25 carriers, $P = 0.32$). For aspartate kinase, the differences within Mexico (268 compared with 232, 75 carriers, $P = 6.4 \times 10^{-4}$) and within Central America (445 compared with 403, 39 carriers, $P = 0.06$) were smaller, indicating that part of its overall association reflects the higher soil N of Central America, where the allele is more common, although carriers still came from N-richer soils within Mexico. Thus, the minor alleles at both loci are more frequent at N-rich collection sites, most clearly for ω-amidase within Mexico, although the regional concentration of carriers means that shared ancestry cannot be excluded. S8_138640842, the strongest of the 12 SNPs that passed both tests ($P = 3.5 \times 10^{-11}$), lies within *Zm00001eb356170* (methionine S-methyltransferase). However, it was supported only by BLINK and FarmCPU and was inconclusive in the correlation test. At 5–15 cm, S8_133179205 lies 0.1 kb from *Zm00001eb354640* (S-adenosylmethionine synthase) and was not

Table 2 Soil-N SNPs detected by at least two GWAS models that passed both environmental specificity tests. SNPs were FDR-significant (5%) for soil N at 0–5 cm in at least two of MLM (M), BLINK (B) and FarmCPU (F). They were not significant in the GWAS of latitude, longitude, elevation, soil pH, CEC or aridity (Supplemental Table S11), and their genotype was not in the top 1% of the genome for correlation with any of those variables (Supplemental Table S12). P, lowest P across models. 5–15 cm, also FDR-significant for soil N at 5–15 cm. Corr., genotype–environment call, either not linked (genotype in the top 5% for correlation with soil N and not in the top 1% for any other variable) or inconclusive (genotype outside the top 5% for correlation with soil N, so the association is detected only by the multi-locus models). Gene (distance), gene within 25 kb of the SNP and its distance, or within if the SNP lies in the gene.

SNP	Chr	Position	Models	MAF	P	5–15 cm	Corr.	Gene (distance)	Annotation
S8_138640842	8	138,640,842	BF	0.254	3.5×10^{-11}	no	inconclusive	*Zm00001eb356170* (within)	Methionine S-methyltransferase
S1_87643408	1	87,643,408	BF	0.356	2.6×10^{-10}	no	inconclusive	*Zm00001eb022740* (within)	Syndetin
S2_191754648	2	191,754,648	BF	0.074	6.2×10^{-10}	no	inconclusive	*Zm00001eb100890* (within)	Uncharacterized protein
S5_22338979	5	22,338,979	MBF	0.030	1.9×10^{-8}	yes	not linked	*Zm00001eb220040* (4.4 kb)	Amino acid kinase domain
S8_153826994	8	153,826,994	BF	0.020	5.1×10^{-8}	no	inconclusive	*Zm00001eb359990* (within)	Homeobox-DDT protein RLT1
S3_54622531	3	54,622,531	BF	0.050	6.5×10^{-8}	yes	not linked	*Zm00001eb130210* (17.9 kb)	RING/FYVE/PHD zinc finger (TTL1-like)
S5_22338938	5	22,338,938	MB	0.031	6.5×10^{-8}	no	not linked	*Zm00001eb220040* (4.3 kb)	Amino acid kinase domain
S1_16128632	1	16,128,632	BF	0.016	2.9×10^{-7}	no	not linked	*Zm00001eb005710* (within)	ω-amidase/NIT2-like CN-hydrolase
S5_53136163	5	53,136,163	BF	0.221	5.7×10^{-7}	yes	inconclusive	*Zm00001eb226000* (within)	Protein kinase
S6_37926580	6	37,926,580	BF	0.179	7.4×10^{-7}	no	inconclusive	*Zm00001eb265710* (within)	HB transcription factor
S1_5543355	1	5,543,355	BF	0.090	1.2×10^{-6}	yes	inconclusive	*Zm00001eb001920* (within)	Transcription repressor
S2_117202525	2	117,202,525	BF	0.071	2.1×10^{-6}	no	not linked	–	No gene within 25 kb

linked to other site variables. The same gene was a candidate in the grain amino acid GWAS (*I.IVL*; $P = 1.0 \times 10^{-8}$). Thus, the soil-N GWAS in landraces, like the grain amino acid GWAS, points to Asp-family and amide nitrogen metabolism.

Free amino acid profiles of Indian Chief and Jarvis show population-specific remodeling of the soluble amino acid pool

The Goodman–Buckler panel captured standing variation in grain amino acids, and landraces captured variation tied to natural soil N gradients, but neither shows how the grain amino acid pool responds to yield selection. The Indian Chief and Jarvis maize populations address this question. Both were base populations in a 14-cycle reciprocal recurrent-selection experiment for yield, and selection shifted N allocation differently in seeds and stover (Moll et al., 1994). Under nonlimiting N, Jarvis Cycle 14 increased grain N mainly via greater grain weight, driven largely by higher post-anthesis soil N uptake, with little change in apparent stover-to-grain N remobilization. Indian Chief Cycle 14 maintained more post-anthesis stover biomass, lost less stover N after anthesis, and did not significantly increase grain N (Moll et al., 1994). Thus, grain N gains can arise from improved soil N recovery or altered internal N allocation during grain filling (Mueller et al., 2019). Because amino acids deliver N to the kernel, we asked whether yield selection also remodeled the grain free amino acid pool, and whether populations with different N responses changed differently. We therefore treat the Indian Chief and Jarvis amino acid profiles as evidence of changes in the grain amino acid pool, not of changes in N allocation or remobilization.

We compared total FAA, the fractions of Pro, Asn, Asp, Gln, and Glu in the free pool, the N-rich FAAs (Asn + Gln + Arg + Lys + His), and FAA-derived N between Cycle 0 and Cycle 14 within each population, correcting for these eight tests (Figure 3; Supplemental Table S14). We also tested all 20 FAA fractions and the protein-bound pool separately (Supplemental Tables S15–S17). Indian Chief showed significant changes in total FAA and composition. Total FAA per mg increased from 1.40 (Cycle 0) to 2.46 (Cycle 14) (Wilcoxon $P = 0.0043$; BH $P = 0.017$; Figure 3A), with no overlap between cycles, and FAA-derived N increased accordingly (1.75 to 2.99 per mg; BH $P = 0.017$). Pro increased from 0.320 to 0.519 of total FAA ($P = 0.016$; BH $P = 0.025$), whereas Asp decreased from 0.115 to 0.075 ($P = 0.016$; BH $P = 0.025$), and Glu from 0.140 to 0.062 ($P = 0.0068$; BH $P = 0.018$). Asn and Gln fractions were unchanged (Figure 3B). When all 20 FAA fractions were tested together, Pro, Asp, Glu, Ile, and Val differed nominally ($P \leq 0.016$), but none remained significant after correction across the 20 tests (BH $P = 0.063$; Supplemental Table S15).

Jarvis showed one nominal change, but the test was underpowered with only three Cycle 0 samples. The Asp fraction increased from 0.075 (Cycle 0) to 0.137 (Cycle 14; $P = 0.019$), but it was not significant after correction (BH-adjusted $P = 0.15$). Total FAA per mg, the Pro and Glu fractions, the N-rich FAA pool, and FAA-derived N did not differ between cycles (Supplemental Table S14). The lack of corrected differences in Jarvis may reflect limited sampling, a stable FAA pool, or both. A stable pool is also consistent with Moll et al. (1994), in which the greater grain N of Jarvis Cycle 14 came mainly from greater grain weight and post-anthesis N uptake rather than from increased remobilization from the stover.

To test whether changes in the free pool extended to total grain amino acid N, we converted FAA and PBAA abundances to amino acid derived N per mg. Protein-bound composition did not change across cycles in either population (Supplemental Figure S9; Supplemental Tables S15–S16). In Indian Chief, FAA-derived N increased from 1.75 to 2.99 per mg, but PBAA-derived N was more than 100-fold larger (332 and 364 per mg) and did not change, nor did total amino acid derived N ($P = 0.42$ for both; Supplemental Figure S10; Supplemental Table S17). Because FAAs carry less than 1% of grain amino acid N, the larger free pool in Cycle 14 did not measurably change grain amino acid N, consistent with unchanged grain N in Indian Chief (Moll et al., 1994). Jarvis showed no change in any amino acid derived N pool. Thus, after 14 cycles of selection, Indian Chief had a greater total FAA and a composition that shifted from Asp/Glu toward Pro, whereas Jarvis showed no significant change. The protein-bound pool, which contains most grain amino acid N, was unchanged in both populations. These population-specific responses are consistent with reported differences in N allocation (Moll et al., 1994).

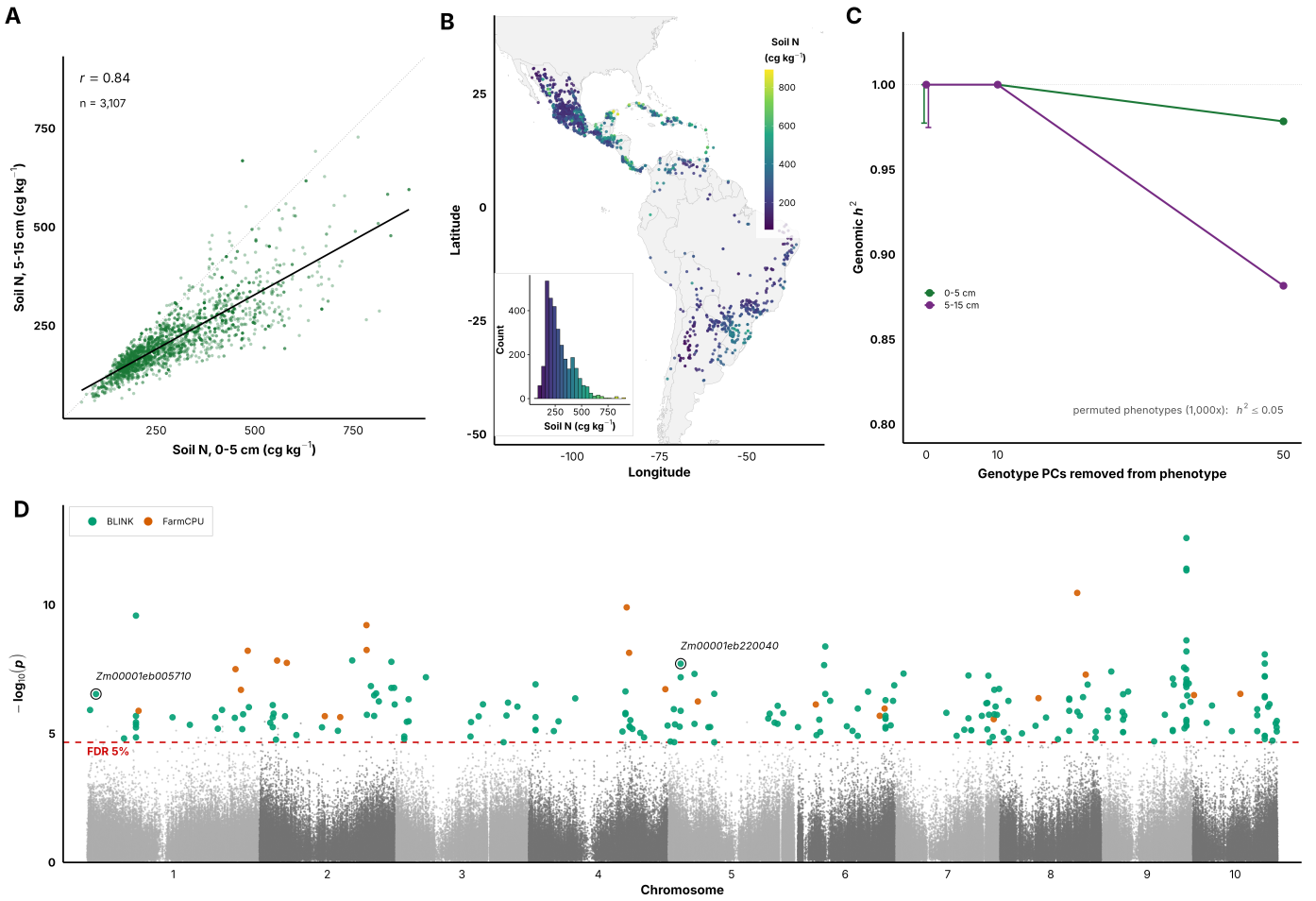


Figure 2 Soil nitrogen at the collection site of 3,107 maize landraces. (A) Soil N at 0–5 cm vs. 5–15 cm ($r = 0.84$); solid line, least-squares fit; dotted line, $y = x$. **(B)** Collection sites colored by soil N at 0–5 cm; inset, distribution of soil N across accessions. **(C)** Genomic heritability (h^2) of soil N at both depths after removing 0, 10 and 50 genotype principal components from the phenotype; error bars, 95% CI at 0 PCs; permuted phenotypes (1,000×) gave $h^2 \leq 0.05$. **(D)** Manhattan plots for soil N (0–5 cm); colored points are SNPs significant at 5% Benjamini–Hochberg FDR in at least one model (MLM, BLINK, FarmCPU; 3 PCs), colored by the model with the lowest P; the SNPs near the aspartate kinase isozyme (*Zm00001eb220040*) and the ω-amidase (*Zm00001eb005710*) are ringed and labeled. The dashed red line marks the 5% FDR threshold. Candidate genes named in the text are labeled.

Skim sequencing reveals that Indian Chief and Jarvis diverged during reciprocal recurrent selection

We skim sequenced 379 individuals from the Indian Chief and Jarvis populations at Cycle 0 and Cycle 14 (Indian Chief 96 and 92; Jarvis 96 and 95). The mean sequencing depth was 0.73× per individual and was higher at Cycle 14 than at Cycle 0 in both populations (Indian Chief 0.61× compared with 0.88×; Jarvis 0.68× compared with 0.76×; Supplemental Figure S11). At this depth, about half of the genotypes in each individual could not be called, and because the number of calls depends on depth, hard-called genotypes would differ between cycles for technical rather than biological reasons. We therefore used genotype likelihoods throughout, estimating population structure from 512,667 filtered SNPs (Supplemental Figure S12) and F_{ST} directly from the aligned reads with ANGSD (Korneliussen et al., 2014).

Unsupervised ancestry estimation ($K = 2$–5) on sites thinned to one per 10 kb separated the four population cycle groups (Figure 4A). Likelihood increased strongly to $K = 4$, where runs converged and each group formed a distinct cluster. However, at $K = 5$, runs disagreed, and likelihood gains were small. At $K = 2$–3, Indian Chief Cycle 0 individuals were inferred as a mixture of the Jarvis Cycle 0 and Indian Chief Cycle 14 clusters, placing the ancestral Indian Chief population between Jarvis and its selected descendant. Genotype-likelihood PCA showed the same pattern (Figure 4B). PC1 (14.3%) separated Indian Chief Cycle 14 from Jarvis Cycle 14, with both Cycle 0 groups near the center, and PC2 (6.7%) separated Cycle 14 from Cycle 0. Overall, the populations, initially less differentiated at Cycle 0, diverged in opposite directions over 14 cycles.

Genetic differentiation confirmed the divergence between the populations. F_{ST} between Indian Chief and Jarvis nearly doubled from 0.111 at Cycle 0 to 0.211 at Cycle 14, increasing across all ten chromosomes (Figure 4C). Because Cycle 14 had deeper sequencing, which should improve allele-frequency estimates and reduce F_{ST}, the increase is unlikely to be a depth artifact. This divergence is expected under reciprocal recurrent selection, which favors complementary rather than shared alleles (Comstock et al., 1949) and is reinforced by drift in small, separately maintained populations. Thus, Indian

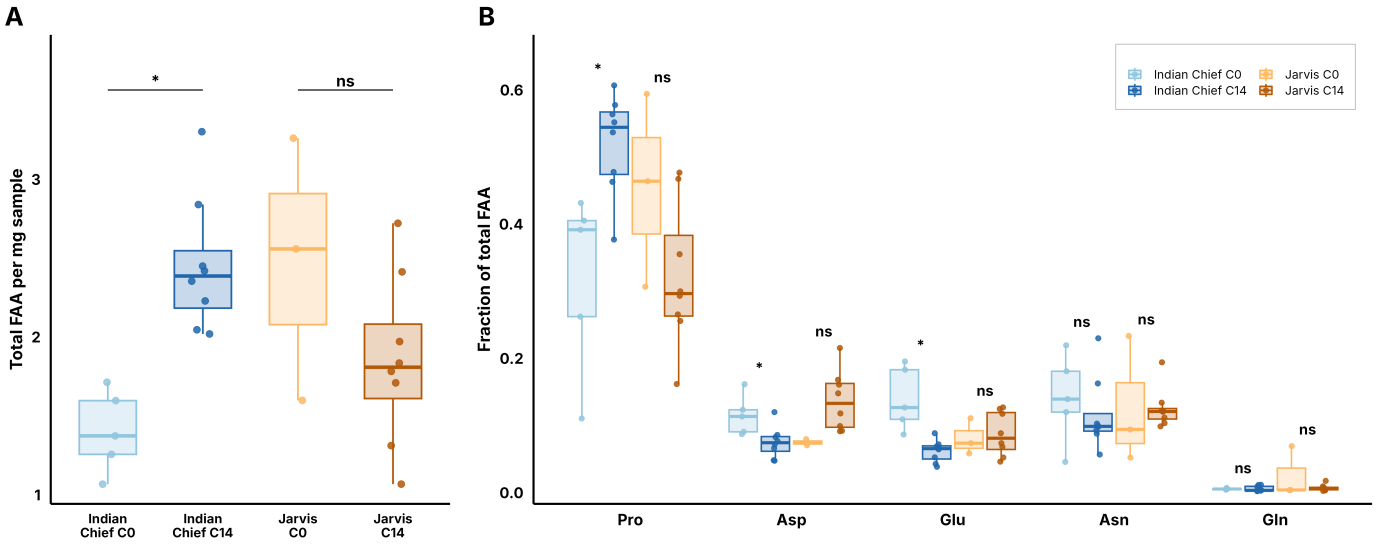


Figure 3 Free amino acid (FAA) pool and composition in Indian Chief and Jarvis before (C0) and after (C14) 14 cycles of reciprocal recurrent selection. **(A)** Total FAA per mg sample. **(B)** Fractions of total FAA for Pro, Asp, Glu, Asn and Gln. Points are biological samples, each the mean of three technical replicates; boxes show the median and interquartile range. Within each population, C0 and C14 were compared with two-sided Wilcoxon rank-sum tests, and P values were adjusted with the BH method; * adjusted $P < 0.05$; ns, not significant.

Chief and Jarvis became more differentiated from each other over 14 cycles of reciprocal recurrent selection.

F_{ST} scans identify population-specific candidate regions in Indian Chief and Jarvis

Before scanning for regions of high differentiation between cycles, we asked how much of the allele frequency change could be explained by genetic drift. In the reciprocal recurrent selection program, each cycle was formed by intermating the self-pollinated progenies of the 20 best plants per population (Moll and Stuber, 1971), which implies an effective population size (N_e) of at most 20 per cycle. Genome-wide mean F_{ST} between Cycle 0 and Cycle 14 was 0.095 in Indian Chief and 0.114 in Jarvis (Figure 5), values reproduced by neutral Wright–Fisher simulations at $N_e \approx 30$ and $N_e \approx 26$ per cycle, respectively (Supplemental Figure S13). Although slightly above the design-based expectation, these estimates indicate that most genome-wide change between cycles is consistent with drift in small populations. A similar conclusion was reached for the Iowa Stiff Stalk Synthetic (BSSS) and Iowa Corn Borer Synthetic No. 1 (BSCB1), two maize populations under reciprocal recurrent selection for hybrid yield, where linked drift simulations from founder haplotypes explained most of the loss of diversity, and few regions deviated from drift or were shared between populations (Gerke et al., 2015). At this N_e, drift moves long linked regions together, so individual outlier regions cannot be formally distinguished from drift. We therefore treat the scans below as exploratory.

We scanned each population for the regions with the largest allele-frequency change between Cycle 0 and Cycle 14 using F_{ST} summarized in windows (2,162 windows in Indian Chief and 2,151 in Jarvis; median length 500 kb). We retained the top 5% of windows ($F_{ST} \geq 0.157$ in Indian Chief and ≥ 0.178 in Jarvis; 109 and 108 windows), which contained 2,001 and 1,179 genes, respectively (Figure 5; Supplemental Tables S18–S21). Candidate regions were largely population-specific. Only five regions (2.5 Mb; 73 genes) were in the top 5% in both populations, no more than expected by chance ($P = 0.80$; Figure 5), indicating that the two populations show no evidence of shared targets of differentiation. Likewise, 192 of the 2,152 Goodman–Buckler amino acid GWAS loci (8.9%) fell in a top-5% window of either population, the same proportion as the genome covered by these windows (8.9%; $P = 0.52$). Thus, the F_{ST} scans and the amino acid GWAS do not converge on the same genomic regions beyond chance.

Each candidate set contains more than 1,000 genes, so genes from almost any metabolic pathway are expected to be present by chance. We therefore focused on genes in the pathways highlighted by the amino acid and soil N GWAS (Asp-family and Glu/Gln metabolism, amino acid and N transport, and protein turnover) and described them as candidates of interest rather than as evidence of selection. Pathway reoccurence across analyses more than expected is tested in the next section. In Indian Chief, candidate regions included genes of the Asp-family amino acid metabolism, *Zm00001eb214360* (aspartate-semialdehyde dehydrogenase) and *Zm00001eb294790* (threonine synthase), and genes of Glu/Gln interconversion, including two glutamate dehydrogenases (*Zm00001eb060520*, *Zm00001eb162300*), a glutamine synthetase domain-containing protein (*Zm00001eb162900*), and a glutamine amidotransferase domain-containing protein (*Zm00001eb217220*), which also contains a significant SNP for proline in the Goodman–Buckler GWAS (Table 1). Candidate regions also contained genes involved in N transport and protein turnover, including the senescence-specific cysteine protease *SAG39* (*Zm00001eb214780*) and an ammonium transporter (*Zm00001eb214890*), which lie in the same window on chromosome 5; an NRT1/PTR-family transporter (*Zm00001eb287950*) and the autophagy-related gene *ATG18g* (*Zm00001eb287690*), which share a window on chromosome 6, as well as two further autophagy-related genes (*Zm00001eb352300* and *Zm00001eb372060*). These

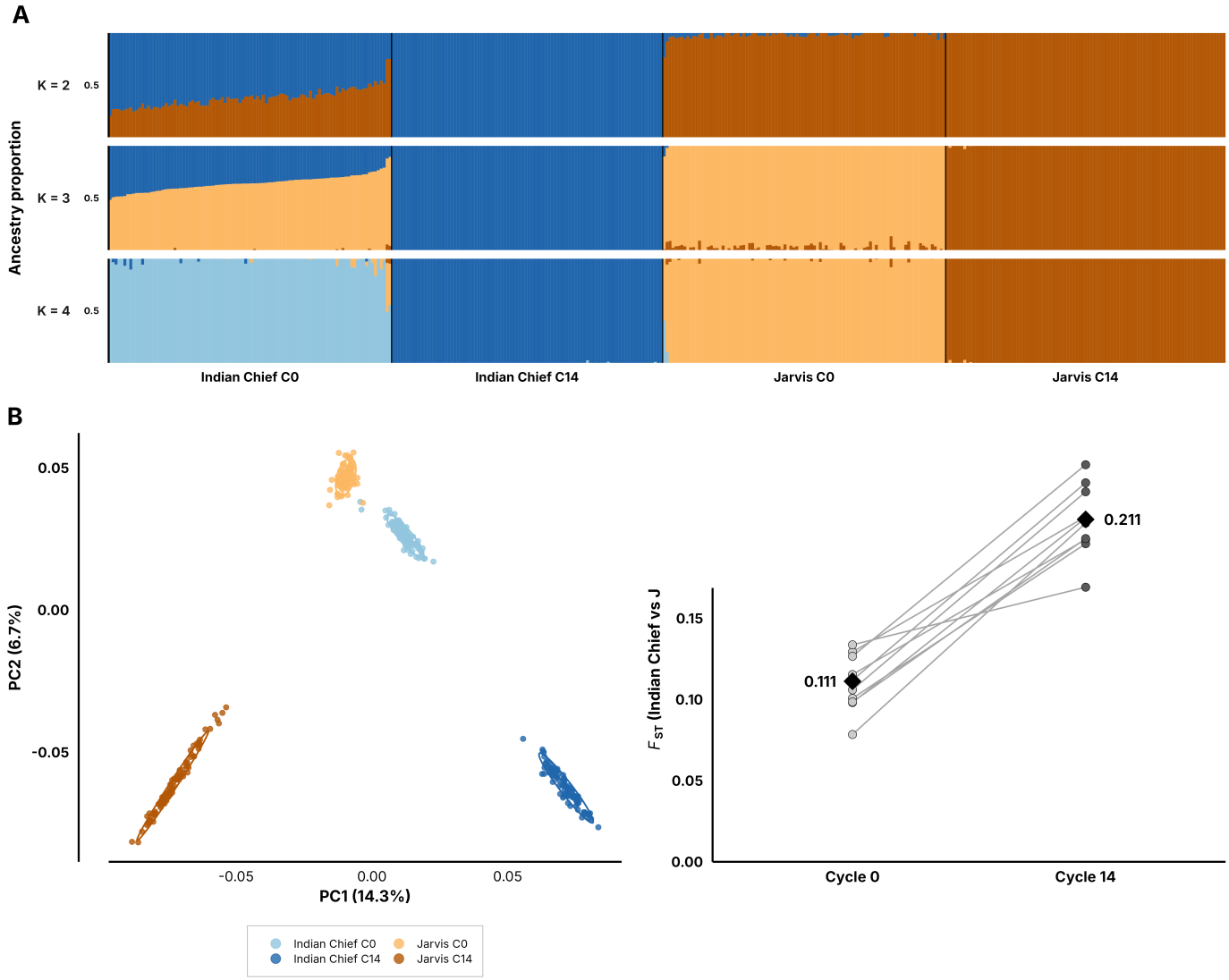


Figure 4 Indian Chief and Jarvis divergence during 14 cycles of reciprocal recurrent selection. All panels use genotype likelihoods. **(A)** Unsupervised ancestry proportions for $K = 2$–4 (thinned to one per 10 kb); each bar is one individual, and each cluster is colored by the group with the highest mean membership. $K = 4$ is the largest K at which independent runs converged. **(B)** Principal component analysis of the genotype-likelihood covariance matrix; ellipses enclose 90% of each group. Individuals with lower depth carry less information and lie closer to the center of their group. **(C)** F_{ST} between Indian Chief and Jarvis at Cycle 0 and Cycle 14, estimated from the joint site-frequency spectrum of each pair; gray lines connect the same chromosome and black diamonds give the genome-wide value.

annotations are compatible with the Indian Chief FAA phenotype, in which total FAA and the Pro fraction increased while the Asp and Glu fractions decreased from Cycle 0 to Cycle 14, but they do not establish that these genes underlie it.

In Jarvis, candidate regions included two genes of Asp metabolism, including *Zm00001eb094670*, annotated as aspartate kinase, and an aspartate aminotransferase (*Zm00001eb152450*, GOT1/THP3). These genes are of interest because the Asp fraction was the only FAA trait that changed in Jarvis, increasing nominally from Cycle 0 to Cycle 14. Jarvis candidate regions also contained two amino acid transporter genes (*Zm00001eb132750*, an amino acid transporter transmembrane domain-containing protein, and *Zm00001eb159580*, a putative GABA transporter), NPL4-like protein 1 (*Zm00001eb146560*), which acts in ubiquitin-dependent protein degradation, and two glutaredoxins (*Zm00001eb131830* and *Zm00001eb131840*). As in Indian Chief, these candidates point to amino acid metabolism, transport, and protein turnover as processes worth testing, but they require functional validation.

The aspartate-to-homoserine pathway recurs across the three populations

To test whether the three populations implicate the same metabolic pathways, we asked whether any CornCyc pathway (Hawkins et al., 2021) contained candidate genes from more than one of the four analyses (amino acid GWAS, soil N GWAS, and the Indian Chief and Jarvis F_{ST} scans) more often than expected by chance. We represented each candidate set by its genomic intervals and generated a null distribution by randomly shifting the intervals

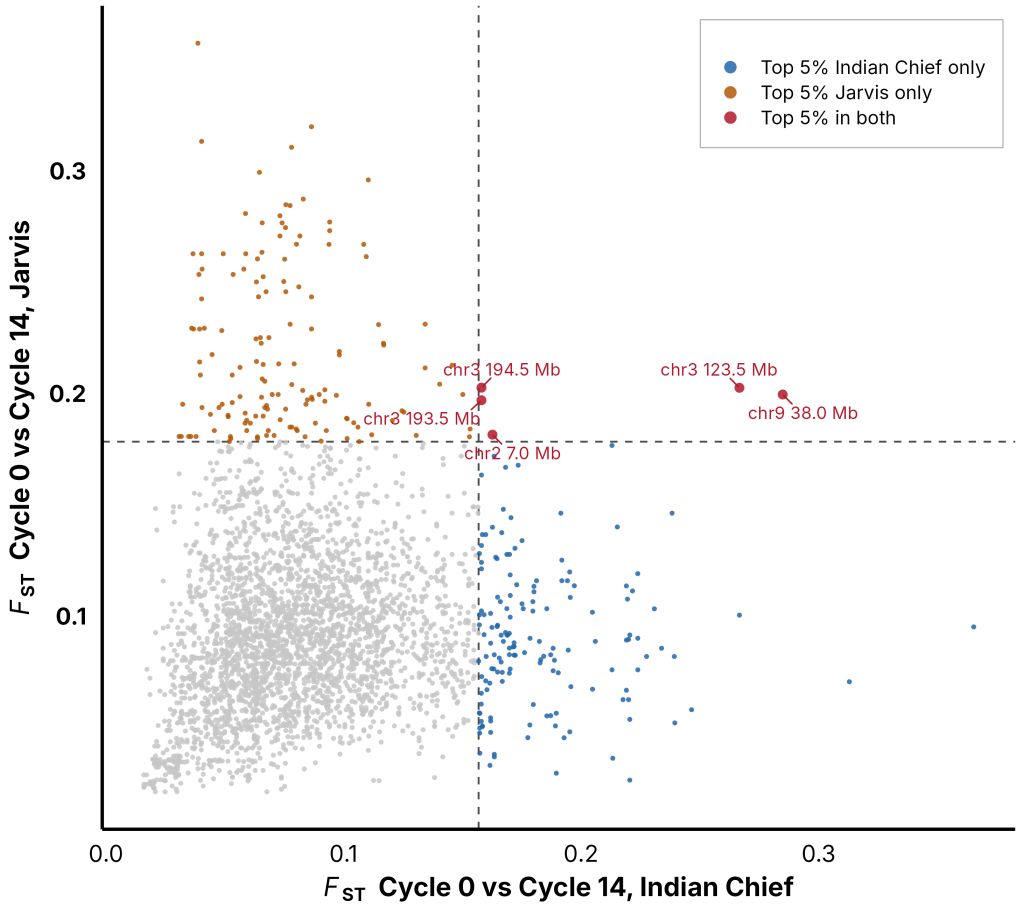


Figure 5 Candidate regions of differentiation between Cycle 0 and Cycle 14 F_{ST} between Cycle 0 and Cycle 14 in Indian Chief (x-axis) and Jarvis (y-axis), estimated from genotype likelihoods and summarized in spline-defined windows. Each point is a genomic segment bounded by the window boundaries of either population, and carries the F_{ST} of the Indian Chief and Jarvis windows that contain it. Dashed lines mark the top 5% of windows in each population ($F_{ST} \geq 0.157$ in Indian Chief and ≥ 0.178 in Jarvis). Blue and orange, top 5% in one population only; red, top 5% in both, labeled with chromosome and position.

along each chromosome, which preserves their number, size, and clustering. Because this test asks whether one pathway stands out from all others, larger candidate sets add background genes that hit many pathways by chance. Therefore, we used stricter sets than in the gene level analyses. The bonferroni significant SNPs for both GWAS and the top 3% of F_{ST} windows. Among 468 pathways, L-homoserine biosynthesis had candidates from all four analyses, had the lowest permutation-based empirical P (P = 0.00034), and was the only pathway that remained significant after correction across all pathways (corrected $P = 0.021$; Figure 6). Each analysis implicated a different enzyme, homoserine dehydrogenase (*Zm00001eb405410*) in the amino acid GWAS (Asn relative to the Asp-family amino acids, *N.IMNTDK*), an aspartate kinase isozyme (*Zm00001eb220040*) in the soil N GWAS, aspartate-semialdehyde dehydrogenase (*Zm00001eb214360*) in Indian Chief, and aspartate kinase (*Zm00001eb094670*) in Jarvis. L-lysine biosynthesis VI also had candidates from all four analyses (P = 0.0030), largely because it shares the aspartate kinase and aspartate-semialdehyde dehydrogenase steps. The other pathways with $P < 0.05$ were small terpenoid and vitamin E pathways with candidates from two or three analyses, and none remained significant after correction.

We reran the test across 12 combinations of GWAS thresholds (FDR 5%, 3%, 1%, and Bonferroni) and F_{ST} window thresholds (top 5%, 3%, 1%) (Supplemental Figure S14). The same four genes were recovered at all GWAS thresholds when using the top 5% or 3% of F_{ST} windows, and in all eight such combinations, L-homoserine biosynthesis ranked first or second among 468 pathways. Across all 12 combinations, it ranked first five times and second three times. The lower ranks occurred only with the top 1% windows (Supplemental Figure S14B). Using larger candidate sets (FDR 5% and top 5% windows), more pathways were hit by chance, and the pathway was not significant after correction (corrected P = 0.68). Thus, L-homoserine biosynthesis ranked consistently across thresholds, while corrected significance depended on candidate-set size. Although candidate regions from the four analyses overlap no more than expected by chance, each implicates a different step in aspartate-to-homoserine conversion.

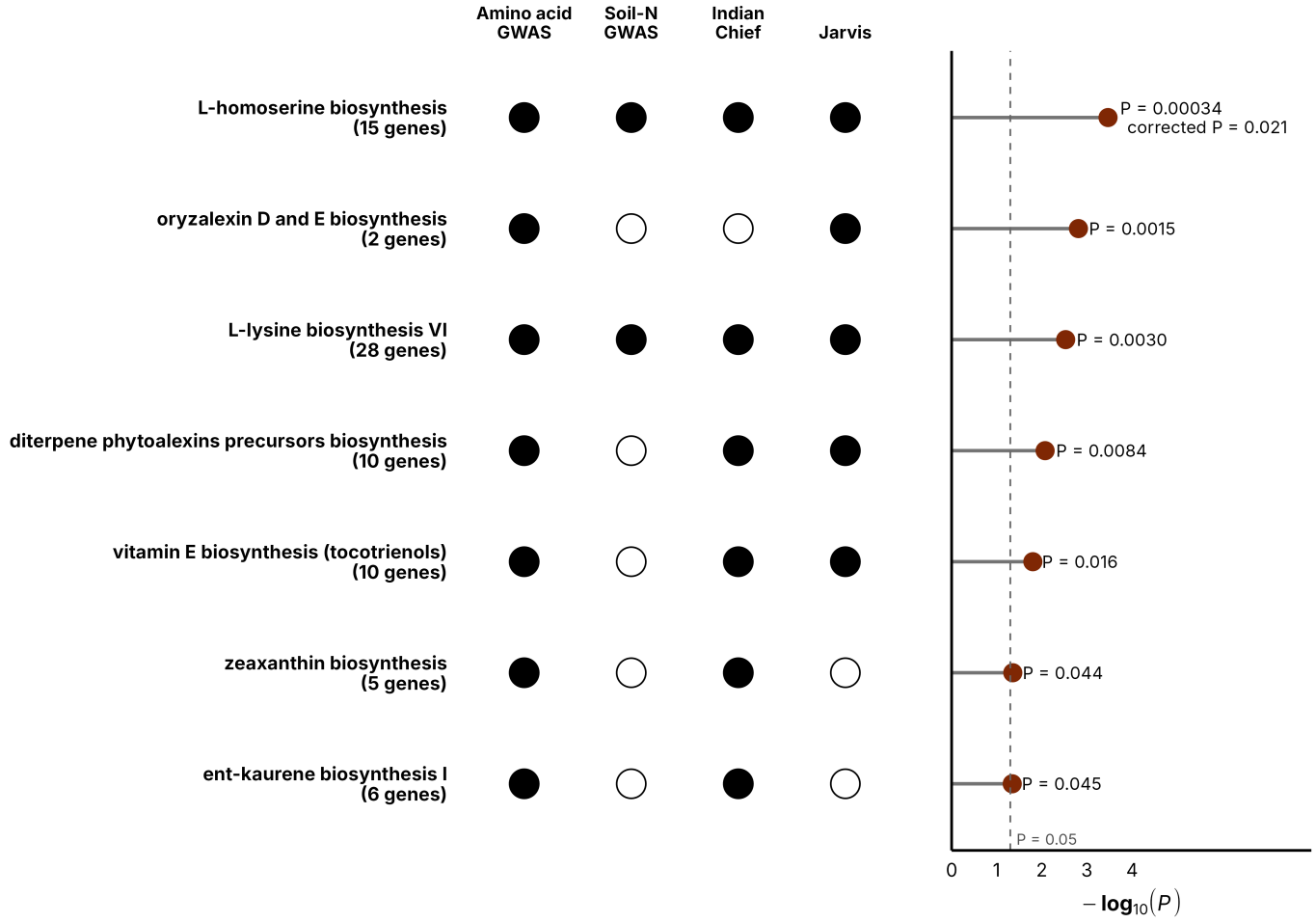


Figure 6 The L-homoserine biosynthesis pathway recurs across all four analyses. All CornCyc pathways with uncorrected permutation $P < 0.05$ for recurrence across the amino acid GWAS, soil-N GWAS and the Indian Chief and Jarvis F_{ST} scans, using Bonferroni-significant GWAS SNPs and the top 3% of F_{ST} windows. Left, filled circles mark analyses with at least one candidate gene in the pathway; numbers in brackets give the number of maize genes in the pathway. Right, pathway-specific permutation-based empirical P for the observed number of analyses hitting the pathway; dashed line, $P = 0.05$. The corrected P (minimum-P correction across all 468 pathways) is shown for the only pathway that remained significant.

Discussion

We tested whether natural variation in grain amino acids, genetic variation linked to soil N, and yield selection implicate the same N-metabolism functions in maize. In the Goodman–Buckler panel, the free amino acid pool was small, dominated by Asn and Pro, highly variable, and heritable. GWAS highlighted candidates for amino acid metabolism, transport, and protein turnover. In landraces, soil N at the collection site associated with SNPs near an aspartate kinase isozyme and an ω-amidase, surviving controls for geographic and climatic confounding. After 14 cycles of reciprocal recurrent selection for yield, Indian Chief and Jarvis differed in how they allocated N, with Jarvis increasing grain N mainly through post-anthesis N uptake and Indian Chief retaining more N in the stover (Moll et al., 1994). In our samples, Indian Chief also accumulated more free amino acids, mostly Pro, and the two populations diverged further genetically. Although most candidates were analysis-specific, each analysis implicated a different enzyme in the aspartate-to-homoserine pathway, the only pathway that remained significant after correction. Next, we analyze multi-trait loci, the aspartate-to-homoserine pathway, and the aspartate–glutamate junction, along with candidate genes involved in transport and protein turnover. These annotation-based candidates require functional validation.

Loci shared by many amino acid traits point to protein turnover and membrane protein biogenesis rather than specific biosynthetic steps

Few GWAS loci were shared across many amino acid phenotypes. Phenotype counts also overstate independent signals because composites and ratios reuse amino acids. The most interpretable case is the chromosome 4 locus at 174.57 Mb, associated with eight phenotypes (including Ala, Val, Lys, Phe, and Thr) spanning four biosynthetic families, and adjacent to *Zm00001eb190800*, encoding a peptidase A1 (pepsin-like aspartic protease) (Figure 1). Aspartic proteases are known to drive bulk protein turnover. In senescing tobacco, chloroplast CND41 degrades Rubisco, and reduced CND41 increases Rubisco retention under N limitation (Kato et al., 2004).

Rubisco breakdown is a major source of amino acids released during senescence (Feller et al., 2008), although A1 proteases also act in development and stress responses (Soares et al., 2019). A single locus affecting amino acids from four biosynthetic families is more easily explained by a change in protein degradation than by a change in one biosynthetic enzyme, which makes *Zm00001eb190800* a plausible candidate for functional testing.

The chromosome 9 locus implicates the endomembrane system. A ∼13 kb region associated with nine phenotypes (including total free amino acids) spans *Zm00001eb394360* and *Zm00001eb394380*. *Zm00001eb394360* encodes ER membrane protein complex subunit 4, which inserts transmembrane proteins into the ER (Guna et al., 2018), and *Zm00001eb394380* encodes rhomboid-like protein 19, an intramembrane serine protease annotated for Golgi-to-ER retrograde transport. Although the functions of most plant rhomboid-like proteins remain unknown (Lavell et al., 2021), both genes act in the ER–Golgi system that processes transporters and secreted proteins involved in amino acid and N movement (Brandizzi, 2018), so either could affect the overall free amino acid pool. Because the associations do not clearly separate these genes from nearby genes (including a DUF2404 membrane protein at the chromosome 4 locus), both loci should be treated as positional candidates, and kernel expression and mutant analyses are the most direct tests. Candidate annotations, trait associations, and model-specific GWAS results are available at `https://j83gk2-nirwan-tandukar.shinyapps.io/amino_gwas_app/`.

The aspartate-to-homoserine pathway links soil-N adaptation, recurrent selection, and grain amino acid variation

Asn was the most abundant free amino acid in the Goodman–Buckler panel and, because of its high N-to-C ratio, is a major N transport and storage form in plants (Lea et al., 2007). It is synthesized from Asp by asparagine synthetase using the amide N of Gln, and catabolism returns carbon to Asp or oxaloacetate (Lea et al., 2007). Asp is thus both the precursor of Asn and the entry point to the Asp-family pathway that produces Lys, Thr, Met, and Ile (Han et al., 2021; Wang et al., 2018; Viola, 2001). Pathway flux is regulated mainly by feedback inhibition of aspartate kinase by Lys and Thr, and of dihydrodipicolinate synthase by Lys (Azevedo et al., 2006; Jander and Joshi, 2009). This pathway was implicated by all four analyses. L-homoserine biosynthesis had a candidate gene in every analysis, had the lowest permutation-based empirical P of the 468 CornCyc pathways, and was the only pathway that remained significant after correction (Figure 6).

Across analyses, implicated genes spanned the Asp to homoserine pathway. Jarvis regions included aspartate kinase (*Zm00001eb094670*), the first committed step (Viola, 2001), and *ask2* expression is linked to Met accumulation in developing kernels (Duo et al., 2026). The only soil N SNP significant in all three GWAS models, which also passed both tests for confounding with geography and climate, was near another aspartate kinase (*Zm00001eb220040*). Indian Chief regions included aspartate-semialdehyde dehydrogenase (*Zm00001eb214360*), and the amino acid GWAS implicated homoserine dehydrogenase (*Zm00001eb405410*), the first committed step to Thr and Met. Further downstream, Indian Chief regions included threonine synthase (*Zm00001eb294790*) and methionine synthase (*Zm00001eb052750*), the latter also near a soil-N SNP. The amino acid GWAS also identified genes in the Lys branch, dihydrodipicolinate synthase (*Zm00001eb389290*), dihydrodipicolinate reductase (*Zm00001eb176170*), and diaminopimelate decarboxylase (*Zm00001eb406820*), and in the Asn branch, two asparagine synthetase domain genes (*Zm00001eb036060*, *Zm00001eb036070*) and the amino acid transporter AVT6C (*Zm00001eb368680*), suggesting that transport or storage, not only synthesis, may shape the Asn pool.

Together, three populations with different germplasm and study designs point to the same short pathway, with each analysis implicating a different enzyme. This pattern did not depend on a single threshold. Candidate thresholds are conventions, and across GWAS and F_{ST} thresholds, the same four genes were recovered, with L-homoserine biosynthesis ranked first or second among 468 pathways. It became significant after correction when stricter sets reduced the number of background genes (Supplemental Figure S14), so its consistent ranking is more informative than any single P value. Drift could still contribute to the F_{ST} candidate regions given $N_e \approx 25$–30 (Supplemental Figure S13), and soil N candidates may reflect local adaptation even though they passed tests for geography and climate. For the two soil-N candidates, the minor alleles came from many independent collection sites and were more frequent at N-rich sites, most clearly for ω-amidase within Mexico (Supplemental Figure S8), so neither signal rests on a single site, although shared ancestry may contribute.

Therefore, these genes are strong candidates for direct tests, such as comparing Asp-family amino acids among lines with alternative alleles at *Zm00001eb405410*, *Zm00001eb220040*, *Zm00001eb214360*, and *Zm00001eb094670*. Aspartate is a branch point in plant N metabolism. It is the precursor of Asn, a major transport and storage form of N (Lea et al., 2007). It also exchanges amino groups with Glu via transamination, and it is synthesized from the TCA cycle intermediate oxaloacetate, linking amino-acid N to carbon metabolism (Han et al., 2021). Variation at the entry to the Asp-family pathway could shift how Asp-derived N is allocated among Asn, Asp-family amino acids, and the Glu/Gln pool, consistent with the association of an aspartate kinase allele with soil N in landraces. Links from Asp to Glu/Gln and the ω-amidase pathway are discussed next.

Candidate genes at the aspartate–glutamate junction link amino acid N with C–N recycling

Glutamate exchanges amino groups with the TCA-cycle intermediate 2-oxoglutarate, which provides the carbon skeleton for ammonium assimilation through the GS/GOGAT cycle and glutamate dehydrogenase, and whose level signals C and N status in many organisms (Huergo and Dixon, 2015). Together, Asp, Asn, Glu, and Gln make up most of the free amino acid pool and are major N transport forms in plants (McAllister et al., 2016). Because no Glu/Gln pathway recurred across analyses, we discuss these junction candidates individually.

Each analysis implicated different genes at the junction. On the Asp side, Jarvis regions contained the aspartate aminotransferase *Zm00001eb152450* (GOT1), which interconverts Asp and 2-oxoglutarate with oxaloacetate and Glu (Han et al., 2021; Zhou et al., 2009). This gene was recently identified as *THP3*, whose teosinte alleles improve N assimilation and seed protein in maize (Huang et al., 2026). In contrast, overexpressing a cytosolic aspartate

aminotransferase from alfalfa in *Brassica napus* did not increase N use efficiency (McAllister et al., 2016), suggesting that the effects of this enzyme depend on the isoform and the tissue in which it is expressed. The soil-N GWAS identified an ω-amidase (*Zm00001eb005710*) that completes Asn/Gln transaminase routes by returning carbon to oxaloacetate and 2-oxoglutarate while releasing ammonia (Zhang and Marsolais, 2014).

On the Glu side, the clearest case is Indian Chief, the only population whose free amino acid profile changed (Figure 3). Most of the Cycle 14 increase in total free amino acids was Pro, so the lower Asp and Glu fractions largely reflect Pro accumulation. Indian Chief regions included two glutamate dehydrogenases (*Zm00001eb060520*, *Zm00001eb162300*), a glutamine synthetase (*Zm00001eb162900*), and a class I glutamine amidotransferase (*Zm00001eb217220*) that was also a candidate for free Pro in the amino acid GWAS, but no genes for Pro synthesis. GDH mutants show altered amino acid profiles under carbon limitation (Miyashita and Good, 2008; Dellero, 2020), and cereal GS supplies Gln to developing grain (Wei et al., 2021; Fujita et al., 2022). Because Pro is synthesized from Glu (**??**), increased flux from Glu to Pro, rather than increased primary N assimilation, is a plausible explanation, although we did not measure flux or enzyme activity. Glutamine amidotransferases function broadly, so *Zm00001eb217220* cannot be assigned to the Glu/Gln cycle without functional evidence, although rice *FLOURY ENDOSPERM19* affects grain filling and C–N metabolism (Lou et al., 2021). Because most grain N is protein-bound, these shifts changed the soluble pool without affecting total grain amino acid N. Jarvis showed no corrected change, consistent with Moll et al. (1994), in which its grain N gain came mainly from higher grain weight and post-anthesis N uptake. In the amino acid GWAS, GS/GOGAT genes (*Zm00001eb329710*, *Zm00001eb360480*, *Zm00001eb059870*) were associated with ratio traits, including Asn/Glu. Thus, all three populations implicated genes at the junction where Asp and Glu exchange N with TCA-cycle carbon. These genes differ among analyses and lack pathway-level support, but finding the same junction in three independent designs makes it a priority for testing how maize balances amino acid N against carbon supply.

Amino acid transport and protein turnover candidates

Beyond Asp and Glu metabolism, each analysis contained candidates for amino acid transport or protein turnover, and these differed among populations in ways that matched their phenotypes. They are individual candidates rather than recurring pathways, and because F_{ST} windows in small populations can reflect drift (Gerke et al., 2015), we considered Indian Chief and Jarvis candidates only when they matched the amino acid phenotypes.

Protein-turnover candidates were most relevant in Indian Chief, where Cycle 14 grain had higher free amino acids but unchanged protein-bound amino acids. Candidate regions included *Zm00001eb352300*, homologous to Arabidopsis *ATG18a* required for autophagosome formation during nutrient stress and senescence (Xiong et al., 2005), and the cysteine protease *SAG39* (*Zm00001eb214780*). Its Arabidopsis homolog *SAG12* localizes to senescence-associated vacuoles (Otegui et al., 2005) and supports N allocation to seeds (James et al., 2018). The same window also contained *Zm00001eb214890*, homologous to the ammonium transporter *AtAMT2* (Neuhäuser et al., 2009). Increased protease or autophagy activity in Cycle 14 could expand the free amino acid pool, but these activities were not measured.

Jarvis candidate regions instead contained genes involved in amino acid import and N signaling. An LHT-like gene (*Zm00001eb132750*) encodes a plasma membrane amino acid importer. In Arabidopsis, LHT1 imports amino acids into root and leaf cells, reducing losses (Fan et al., 2023). A ProT-like gene (*Zm00001eb159580*) transports Pro, glycine betaine, and GABA between cells (Lehmann et al., 2011). These regions also included a tandem pair of CC-type glutaredoxin genes (*Zm00001eb131830*, *Zm00001eb131840*) in two highly differentiated windows (ranked 2nd and 7th out of 2,151). In a reanalysis of public maize leaf RNA-seq data (Mu et al., 2017; Guo et al., 2022; Ma et al., 2021; Torres-Rodríguez et al., 2021; Alonso-Nieves et al., 2022; Wang et al., 2025), both genes were strongly downregulated under low nitrate (*Zm00001eb131840*; about 80-fold lower), although their expression in developing kernels is unknown (Supplemental Table S22). In Arabidopsis, CC-type glutaredoxins function in N signaling, including nitrate-induced regulators of root growth and the mobile peptides CEPD1/CEPD2 that increase nitrate uptake under N demand (Patterson et al., 2016; Ohkubo et al., 2017). Both glutaredoxin genes and the LHT-like gene also lie within the chromosome 3 Asp and Pro QTL interval of the B64 × CML14 population (Supplemental Figure S5). Because Jarvis increased grain N mainly via post-anthesis N uptake (Moll et al., 1994), amino acid import and N-signaling genes may be more relevant than changes in grain amino acid composition. Jarvis regions also contained NPL4-like protein 1 (*Zm00001eb146560*). Its Arabidopsis homolog partners with UFD1 and CDC48 to target ubiquitinated proteins (including Rubisco large subunit) for proteasomal degradation (**?**).

In the amino acid GWAS, several transporter candidates mapped to specific free amino acids. Three UmamiT genes (WAT1-related exporters) were associated with Gln (*Zm00001eb379220*), Asp (*Zm00001eb207630*), and Lys (*Zm00001eb352400*). Because seed-coat UmamiTs export maternal amino acids to the embryo and mutants reduce supply (Müller et al., 2015), these associations suggest that variation in export to the grain can alter individual free amino acids without changing synthesis. The NRT1/PTR family gene *NPF4.6* (*Zm00001eb277160*), whose family transports nitrate, peptides, amino acids, and hormones (Léran et al., 2014), was linked to free Lys. The Gln/Glu ratio mapped to the amino acid permease *aaap40* (*Zm00001eb261470*) in a region also containing *LHT1* and *LHT2*, while a cationic amino acid transporter (*Zm00001eb050630*) was associated with branched-chain amino acids. Two loci pointed to vacuolar storage, including an *AVT3A*-like gene (*Zm00001eb429100*) associated with Ile, consistent with Arabidopsis homologs that export vacuolar amino acids (Fujiki et al., 2017), and *AVT6C* (*Zm00001eb368680*) associated with free Asn. Thus, transport and turnover candidates differed across populations in phenotype-consistent ways: protein turnover in Indian Chief; amino acid import and N signaling in Jarvis; and amino acid export and vacuolar storage in the amino acid GWAS. Collectively, these candidates trace N flow to the grain—from uptake and recycling to delivery and storage—suggesting each population varies at a different step.

Conclusion

Across three maize populations, candidate genes for grain amino acid variation fell into three groups. The first is the aspartate-to-homoserine pathway, the only pathway significant after correction, in which each analysis implicated a different entry step, aspartate kinase in Jarvis (*Zm00001eb094670*) and the landrace soil-N GWAS (*Zm00001eb220040*), aspartate-semialdehyde dehydrogenase in Indian Chief (*Zm00001eb214360*), and homoserine dehydrogenase in the Goodman–Buckler panel (*Zm00001eb405410*). These steps may control how Asp-derived N is divided between Asn and the Asp-family amino acids. The second is the junction where Asp and Glu exchange N with TCA-cycle carbon, with aspartate aminotransferase in Jarvis, ω-amidase in the landraces, and glutamate dehydrogenase and glutamine synthetase in Indian Chief. The third is amino acid transport and protein turnover, with a peptidase A1 gene (*Zm00001eb190800*) at a Goodman–Buckler locus affecting amino acids from four biosynthetic families, senescence protease and autophagy candidates in Indian Chief, amino acid importers in Jarvis, and UmamiT exporters and vacuolar AVT transporters linked to specific free amino acids in the Goodman–Buckler panel.

These genes are candidates, not confirmed mechanisms. Pathway recurrence was significant after correction only with stricter candidate sets, drift may underlie the differentiation between cycles, and we did not measure N uptake or remobilization. Because the three groups act sequentially, partitioning N among amino acids, linking amino acid N to carbon supply, and reallocating N between tissues and storage, they can be tested together. For example, by combining alleles of an Asp-pathway gene and an amino acid transporter in the same background. Testing alleles and mutants, along with kernel expression under varying N and ^{15}N labeling to distinguish uptake from remobilization, will identify which steps, alone or combined, are best targets to improve maize N use.

Materials and methods

Full details of all analyses are given in Supplemental Methods.

Plant material

The Goodman–Buckler maize association panel (Flint-Garcia et al., 2005) was grown at the Genetics Farm near Columbia, Missouri, in 2017 and 2018, with two replicates per year in a randomized complete block design, and 280 of the 282 lines were grown successfully. Self-pollinated ears of each line were dried, shelled, and bulked within each replicate and year, and 25 kernels per bulk were ground for amino acid analysis. The B64 × CML14 mapping population was developed by crossing B64 and CML14, the Goodman–Buckler lines with the lowest and highest grain free Asn, respectively, and its $F_{2:3}$ families were grown and sampled as described in Fitzsimmons et al. (2026).

We used 3,107 georeferenced maize landrace accessions from the CIMMYT International Germplasm Bank (Romero Navarro et al., 2017), collected in 28 countries and territories across Mesoamerica (2,296), South America (667), and the Caribbean (144), for which soil N values were available at the collection sites.

We used Cycle 0 and Cycle 14 of two open-pollinated populations, Indian Chief and Jarvis Golden Prolific, from a program of reciprocal recurrent selection for grain yield (Moll and Stuber, 1971; Moll et al., 1994). Parents in each population were selected on the yield of their testcrosses with the other population, so allele-frequency changes in the two populations are not fully independent, although no alleles were exchanged between them. For amino acid profiling, Cycle 0 and Cycle 14 plants of both populations were grown in 2025 at the Central Crops Research Station (Clayton, NC) in a randomized complete block design, and whole cobs with fully developed kernels were harvested.

Phenotyping

PBAAs were quantified after acid hydrolysis by LC-MS/MS (Yobi and Angelovici, 2018). Hydrolysis converts Asn and Gln to Asp and Glu (reported as Asx and Glx) and destroys Trp and Cys, giving 15 PBAAs in the Goodman–Buckler panel and 16 in Indian Chief and Jarvis, where Gly was also quantified. All 20 FAAs were quantified by UPLC-MS/MS (Gu et al., 2007). Amino acid-derived N weighted each amino acid by its number of N atoms, with Asx and Glx assigned 1.5 N. Free and protein-bound pools were compared in the 279 Goodman–Buckler lines with both measurements (Shrestha et al., 2022) by paired Wilcoxon signed-rank tests, combining free Asp with Asn and free Glu with Gln to match Asx and Glx.

For Indian Chief and Jarvis, technical replicates were averaged within each biological sample (Indian Chief $n = 5$ and 8, Jarvis $n = 3$ and 8, for Cycle 0 and Cycle 14). Cycles were compared within each population by two-sided Wilcoxon rank-sum tests, with Benjamini–Hochberg adjustment within each family of tests (Supplemental Tables S14–S17).

Soil total N at 0–5 cm and 5–15 cm was taken from SoilGrids 2.0 (Poggio et al., 2021) at 250-m resolution and assigned to the collection coordinates of each landrace accession. The 0–5 cm layer was the primary GWAS phenotype, and 5–15 cm was used as a check.

Genotyping

The Goodman–Buckler panel was genotyped with the 4,177,796 maize HapMap 3.2.1 SNPs (Bukowski et al., 2018), whose AGPv4 positions were lifted to B73 RefGen_v5 (Zm-B73-REFERENCE-NAM-5.0). F_2 plants of the B64 × CML14 population were genotyped with the Agriplex Genomics mid-density temperate corn panel as described in Fitzsimmons et al. (2026). The landraces were genotyped by genotyping-by-sequencing (Romero Navarro et al., 2017), and we used 495,755 SNPs on RefGen_v5. For Indian Chief and Jarvis, 96 individuals per population–cycle group were skim sequenced (2 × 150 bp reads, NovaSeq 6000), reads were aligned to RefGen_v5 with BWA-MEM, and variants were called with GATK. After five low-quality samples were removed, 379 individuals remained, with a mean depth of 0.73× per individual.

Heritability

Broad-sense heritability (H^2) on a line-mean basis was estimated on the log scale with a linear mixed model including line, year, line × year, and block within year, and genomic heritability (h^2) with a VanRaden genomic relationship matrix for the 272 genotyped lines, with a permuted-phenotype check. Consistency across years was the Spearman correlation between 2017 and 2018 line means for the 251 lines grown in both years. For soil N, genomic heritability was estimated for the 3,107 landraces before and after removing 10 and 50 genotype principal components from the phenotype.

Genome-wide association studies

GWAS of the 78 amino acid phenotypes (Supplemental Table S2) used phenotype BLUPs and the HapMap 3.2.1 SNPs in the HAPPI pipeline with the GAPIT MLM, MLMM, BLINK, and FarmCPU models (Slaten et al., 2020). Significance was set at a 5% Benjamini–Hochberg FDR for each phenotype and model, and MLMM gave no significant SNPs. Candidate genes were RefGen_v5 genes within 25 kb of a significant SNP.

GWAS of soil N used MLM, BLINK, and FarmCPU with three genotype principal components as covariates and a 5% Benjamini–Hochberg FDR for each model. To test for confounding with geography and climate, we ran the same GWAS on latitude, longitude, elevation (Romero Navarro et al., 2017), soil pH, CEC (Poggio et al., 2021), and aridity (Zomer et al., 2022), and compared the overlap of significant SNPs with chance by hypergeometric tests. In a second, model-free test, every SNP was ranked genome-wide by the correlation between its allele dosage and each site variable, and a soil-N SNP was called linked to the environment if it ranked in the top 1% for any non-N variable.

QTL mapping

BLUPs of free Asp and free Pro for the 250 $F_{2:3}$ families were mapped on the 610-marker genetic map of Fitzsimmons et al. (2026) by Haley–Knott regression in R/qtl (Broman et al., 2003), with $\alpha = 0.05$ thresholds from 1,000 permutations and 1.5-LOD support intervals. Genes in the shared interval were taken from the RefGen_v5 annotation (Supplemental Table S8).

Population-genomic analyses

F_{ST} between Cycle 0 and Cycle 14 was estimated in each population from genotype likelihoods with ANGSD (Korneliussen et al., 2014), using a joint site-frequency spectrum estimated with `winsfs` (Rasmussen et al., 2022) as the prior, and summarized in windows with GenWin (Beissinger et al., 2015). Windows in the top 5% of each population were retained as candidate regions. These are empirical outliers, not significance tests. Candidate genes were annotated with PANTHER (Thomas et al., 2022) and Gene Ontology terms (Gene Ontology Consortium et al., 2026). Population structure was inferred with NGSadmix (Skotte et al., 2013) and PCAngsd (Meisner and Albrechtsen, 2018), and differentiation between Indian Chief and Jarvis was measured with Hudson's F_{ST} (Bhatia et al., 2013). N_e was estimated by matching the observed genome-wide F_{ST} between cycles to neutral Wright–Fisher simulations started from the Cycle 0 allele frequencies.

Pathway recurrence

Pathway membership was taken from CornCyc (Plant Metabolic Network release 2023, Hawkins et al., 2021). The main test used Bonferroni-significant GWAS SNPs ±25 kb and the top 3% of windows from each F_{ST} scan. For each of the 468 pathways, we counted how many of the four candidate sets contained a pathway gene and compared this count with 200,000 permutations that circularly shifted each set's intervals along each chromosome, preserving their number, size, and clustering. P values were corrected across pathways with a minimum-P procedure, and the test was repeated for 12 combinations of GWAS and F_{ST} thresholds.

Expression under contrasting N supply

N responses of candidate genes were taken from a reanalysis of published maize RNA-seq studies of root, leaf, shoot, and whole-seedling tissue (Supplemental Table S22), including six leaf N-deficiency studies (Mu et al., 2017; Guo et al., 2022; Ma et al., 2021; Torres-Rodríguez et al., 2021; Alonso-Nieves et al., 2022; Wang et al., 2025). Each study was analyzed with DESeq2, and studies were pooled by random-effects meta-analysis.

Accession numbers

Sequence data from this article can be found in the NCBI Sequence Read Archive under BioProject PRJNAXXXXXX. Gene identifiers refer to the B73 RefGen_v5 annotation (Zm00001eb).

Supplemental data

The following materials are available in the online version of this article.

Supplemental Methods. Detailed methods for plant material, phenotyping, genotyping, heritability, GWAS, QTL mapping, population-genomic analyses, pathway recurrence, and the expression reanalysis.

Supplemental Figure S1. Free amino acid composition of the Goodman–Buckler panel.

Supplemental Figure S2. Heritability of grain amino acid phenotypes in the Goodman–Buckler panel.

Supplemental Figure S3. Free (FAA) and protein-bound (PBAA) amino acid pools across matched Goodman–Buckler lines ($n = 279$).

Supplemental Figure S4. Manhattan plots for a locus on chromosome 9 associated with nine amino acid phenotypes.

Supplemental Figure S5. Free Asp and Pro map to the same QTL on chromosome 3 in the B64 × CML14 population.

Supplemental Figure S6. Overlap of soil-N associations with other environmental GWAS.

Supplemental Figure S7. Genotype–environment correlation test for the soil-N SNPs.

Supplemental Figure S8. Geography and soil N of the two focal soil-N SNPs.

Supplemental Figure S9. Protein-bound amino acid (PBAA) composition in Indian Chief and Jarvis at Cycle 0 (C0) and Cycle 14 (C14).

Supplemental Figure S10. Amino acid-derived nitrogen per mg sample in Indian Chief and Jarvis at Cycle 0 (C0) and Cycle 14 (C14).

Supplemental Figure S11. Per-sample sequencing depth in the Indian Chief and Jarvis skim-sequencing data.

Supplemental Figure S12. Quality summaries for the 512,667 filtered SNPs.

Supplemental Figure S13. Effective population size of the Indian Chief and Jarvis recurrent-selection populations.

Supplemental Figure S14. Sensitivity of the pathway recurrence test to candidate thresholds.

Supplemental Table S1. Free amino acid phenotypes across the Goodman–Buckler Association Panel.

Supplemental Table S2. Amino acid trait definitions.

Supplemental Table S3. Abundance, variability, and heritability of the 20 free amino acids.

Supplemental Table S4. Heritability of free and protein-bound amino acid traits.

Supplemental Table S5. Free vs. protein-bound amino acid pools.

Supplemental Table S6. Amino acid GWAS candidate genes.

Supplemental Table S7. Amino acid GWAS loci.

Supplemental Table S8. Genes in the chromosome 3 Asp/Pro QTL interval of the B64 × CML14 population.

Supplemental Table S9. Soil N GWAS candidate genes.

Supplemental Table S10. Environmental GWAS overlap test.

Supplemental Table S11. Soil-N GWAS SNPs and their overlap with other environmental GWAS.

Supplemental Table S12. Genotype–environment correlation test for soil-N SNPs.

Supplemental Table S13. Expectation test for the genotype–environment correlation.

Supplemental Table S14. Free amino acid metrics in Indian Chief and Jarvis, Cycle 0 vs. Cycle 14.

Supplemental Table S15. Amino acid composition in Indian Chief and Jarvis, Cycle 0 vs. Cycle 14.

Supplemental Table S16. Protein-bound amino acid metrics in Indian Chief and Jarvis, Cycle 0 vs. Cycle 14.

Supplemental Table S17. Amino acid-derived N per mg sample in Indian Chief and Jarvis.

Supplemental Table S18. Indian Chief F_{ST} windows, Cycle 0 vs. Cycle 14.

Supplemental Table S19. Jarvis F_{ST} windows, Cycle 0 vs. Cycle 14.

Supplemental Table S20. Indian Chief candidate genes in top-5% F_{ST} windows.

Supplemental Table S21. Jarvis candidate genes in top-5% F_{ST} windows.

Supplemental Table S22. Expression responses of candidate genes to N supply in public maize RNA-seq.

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Author contributions

TODO.

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Conflict of interest statement

None declared.

Data availability

Raw sequencing reads for the Indian Chief and Jarvis populations are available from the NCBI Sequence Read Archive under BioProject PRJNAXXXXXX, and the filtered SNP set used for population structure analyses is available at Zenodo (doi:10.5281/zenodo.XXXXXXX). Amino acid phenotypes and all Supplemental Data are provided with this article. All code used in this study is available at `https://github.com/nirwan1265/Grain_N_remobilization` (archived at doi:10.5281/zenodo.XXXXXXX). The amino acid GWAS results for the Goodman–Buckler panel can be browsed through a Shiny web application at `https://j83gk2-nirwan-tandukar.shinyapps.io/amino_gwas_app/`

References

A. L. Alonso-Nieves, M. N. Salazar-Vidal, J. V. Torres-Rodríguez, L. M. Pérez-Vázquez, J. A. Massange-Sánchez, C. S. Gillmor, and R. J. H. Sawers. The *pho1;2a'-m1.1* allele of *Phosphate1* conditions misregulation of the phosphorus starvation response in maize (*Zea mays* ssp. *mays* L.). *Plant Direct*, 6(7):e416, 2022. doi: 10.1002/pld3.416.

R. A. Azevedo, M. Lancien, and P. J. Lea. The aspartic acid metabolic pathway, an exciting and essential pathway in plants. *Amino Acids*, 30(2):143–162, 2006.

T. M. Beissinger, G. J. M. Rosa, S. M. Kaeppler, D. Gianola, and N. de Leon. Defining window-boundaries for genomic analyses using smoothing spline techniques. *Genetics Selection Evolution*, 47:30, 2015. doi: 10.1186/s12711-015-0105-9.

S. M. Bernard and D. Z. Habash. The importance of cytosolic glutamine synthetase in nitrogen assimilation and recycling. *New Phytologist*, 182(3):608–620, 2009. doi: 10.1111/j.1469-8137.2009.02823.x.

G. Bhatia, N. Patterson, S. Sankararaman, and A. L. Price. Estimating and interpreting F_{ST}: the impact of rare variants. *Genome Research*, 23(9):1514–1521, 2013. doi: 10.1101/gr.154831.113.

F. Brandizzi. Transport from the endoplasmic reticulum to the Golgi in plants: Where are we now? *Seminars in Cell & Developmental Biology*, 80:94–105, 2018. doi: 10.1016/j.semcdb.2017.06.024.

K. W. Broman, H. Wu, Śaunak Sen, and G. A. Churchill. R/qtl: Qtl mapping in experimental crosses. *Bioinformatics*, 19:889–890, 5 2003. ISSN 1367-4811. doi: 10.1093/bioinformatics/btg112.

R. Bukowski, X. Guo, Y. Lu, C. Zou, B. He, Z. Rong, B. Wang, D. Xu, B. Yang, C. Xie, L. Fan, S. Gao, X. Xu, G. Zhang, Y. Li, Y. Jiao, J. F. Doebley, J. Ross-Ibarra, A. Lorant, V. Buffalo, M. C. Romay, E. S. Buckler, D. Ware, J. Lai, Q. Sun, and Y. Xu. Construction of the third-generation extitZea mays

haplotype map. *GigaScience*, 7(4):gix134, 2018. doi: 10.1093/gigascience/gix134.

M. Carceller, P. Prystupa, and J. H. Lemcoff. Remobilization of proline and other nitrogen compounds from senescing leaves of maize under water stress. *Journal of Agronomy and Crop Science*, 183:61–66, 1999. doi: 10.1046/j.1439-037x.1999.00324.x.

R. E. Comstock, H. F. Robinson, and P. H. Harvey. A breeding procedure designed to make maximum use of both general and specific combining ability. *Agronomy Journal*, 41(8):360–367, 1949.

Y. Dellero. Manipulating amino acid metabolism to improve crop nitrogen use efficiency for a sustainable agriculture. *Frontiers in Plant Science*, 11:602548, 2020. doi: 10.3389/fpls.2020.602548.

H. Duo, R. Chhabra, R. U. Zunjare, G. Sharma, S. J. Mishra, V. Muthusamy, and F. Hossain. Unveiling the link between expression of aspartate kinase2 (ask2) gene and accumulation of methionine in developing maize kernels. *Biochemical Genetics*, 2026. doi: 10.1007/s10528-026-11346-w.

T. Fan, C. Wu, W. Yang, T. Lv, Y. Zhou, and C. Tian. The LHT gene family in rice: Molecular characterization, transport functions and expression analysis. *Plants*, 12(4):817, 2023. doi: 10.3390/plants12040817.

U. Feller, I. Anders, and T. Mae. Rubiscolytics: fate of Rubisco after its enzymatic function in a cell is terminated. *Journal of Experimental Botany*, 59(7):1615–1624, 2008. doi: 10.1093/jxb/erm242.

S. L. Fitzsimmons, V. Shrestha, A. Yobi, R. Angelovici, and S. Flint-Garcia. From kernel to kitchen: Asparagine synthetase 3 is involved in free asparagine accumulation in maize kernels. *bioRxiv*, pages 2026–06, 2026.

S. A. Flint-Garcia, A.-C. Thuillet, J. Yu, G. Pressoir, S. M. Romero, S. E. Mitchell, J. Doebley, S. Kresovich, M. M. Goodman, and E. S. Buckler. Maize association population: a high-resolution platform for quantitative trait locus dissection. *The Plant Journal*, 44(6):1054–1064, 2005. doi: 10.1111/j.1365-313X.2005.02591.x.

Y. Fujiki, H. Teshima, S. Kashiwao, M. Kawano-Kawada, Y. Ohsumi, Y. Kakinuma, and T. Sekito. Functional identification of AtAVT3, a family of vacuolar amino acid transporters, in Arabidopsis. *FEBS Letters*, 591(1):5–15, 2017. doi: 10.1002/1873-3468.12507.

T. Fujita, M. P. Beier, M. Tabuchi-Kobayashi, Y. Hayatsu, H. Nakamura, T. Umetsu-Ohashi, K. Sasaki, K. Ishiyama, E. Murozuka, M. Kojima, H. Sakakibara, Y. Sawa, A. Miyao, T. Hayakawa, T. Yamaya, and S. Kojima. Cytosolic glutamine synthetase GS1;3 is involved in rice grain ripening and germination. *Frontiers in Plant Science*, 13:835835, 2022. doi: 10.3389/fpls.2022.835835.

Gene Ontology Consortium, S. A. Aleksander, et al. The Gene Ontology knowledgebase in 2026. *Nucleic Acids Research*, 54 (D1):D1779–D1792, 2026. doi: 10.1093/nar/gkaf1292.

J. P. Gerke, J. W. Edwards, K. E. Guill, J. Ross-Ibarra, and M. D. McMullen. The genomic impacts of drift and selection for hybrid performance in maize. *Genetics*, 201(3):1201–1211, 2015. doi: 10.1534/genetics.115.182410.

L. Gu, A. D. Jones, and R. L. Last. LC-MS/MS assay for protein amino acids and metabolically related compounds for large-scale screening of metabolic phenotypes. *Analytical Chemistry*, 79(21):8067–8075, 2007. doi: 10.1021/ac070938b.

A. Guna, N. Volkmar, J. C. Christianson, and R. S. Hegde. The ER membrane protein complex is a transmembrane domain insertase. *Science*, 359(6374):470–473, 2018. doi: 10.1126/science.aao3099.

S. Guo, A. Arshad, L. Yang, Y. Qin, X. Mu, and G. Mi. Comparative transcriptome analysis reveals common and developmental stage-specific genes that respond to low nitrogen in maize leaves. *Plants*, 11(12):1550, 2022. doi: 10.3390/plants11121550.

M. Han, C. Zhang, P. Suglo, S. Sun, M. Wang, and T. Su. L-aspartate: An essential metabolite for plant growth and stress acclimation. *Molecules*, 26(7):1887, 2021. doi: 10.3390/molecules26071887.

M. Havé, A. Marmagne, F. Chardon, and C. Masclaux-Daubresse. Nitrogen remobilization during leaf senescence: lessons from Arabidopsis to crops. *Journal of Experimental Botany*, 68 (10):2513–2529, 2017. doi: 10.1093/jxb/erw365.

C. Hawkins, D. Ginzburg, K. Zhao, W. Dwyer, B. Xue, A. Xu, S. Rice, B. Cole, S. Paley, P. Karp, and S. Y. Rhee. Plant metabolic network 15: A resource of genome-wide metabolism databases for 126 plants and algae. *Journal of Integrative Plant Biology*, 63(11):1888–1905, 2021. doi: 10.1111/jipb.13163.

M. Hodges. Enzyme redundancy and the importance of 2-oxoglutarate in plant ammonium assimilation. *Journal of Experimental Botany*, 53(370):905–916, 2002. doi: 10.1093/jexbot/53.370.905.

Y. Huang, Y. Zhu, Y. Cui, G. Shi, X. Wu, W. Li, Z. Chen, Y. Zhou, Y. Gu, Z. Bao, S. Luo, X. Wu, R. Li, J. Liu, X. Dai, J. Liu, D. Chen, L. Gao, C. You, Y. Li, Y. Zhang, W. Wang, H. Wang, and Y. Wu. Teosinte alleles enhance nitrogen assimilation and seed protein in maize. *Nature*, 2026. doi: 10.1038/s41586-026-10575-8.

L. F. Huergo and R. Dixon. The emergence of 2-oxoglutarate as a master regulator metabolite. *Microbiology and Molecular Biology Reviews*, 79(4):419–435, 2015. doi: 10.1128/MMBR.00038-15.

M. James, M. Poret, C. Masclaux-Daubresse, A. Marmagne, L. Coquet, T. Jouenne, P. Chan, J. Trouverie, and P. Etienne. SAG12, a major cysteine protease involved in nitrogen allocation during senescence for seed production in Arabidopsis thaliana. *Plant and Cell Physiology*, 59(10):2052–2063, 2018. doi: 10.1093/pcp/pcy125.

G. Jander and V. Joshi. Aspartate-derived amino acid biosynthesis in *Arabidopsis thaliana*. *The Arabidopsis Book*, 7:e0121, 2009.

Y. Kato, S. Murakami, Y. Yamamoto, H. Chatani, Y. Kondo, T. Nakano, A. Yokota, and F. Sato. The DNA-binding protease, CND41, and the degradation of ribulose-1,5-bisphosphate carboxylase/oxygenase in senescent leaves of tobacco. *Planta*, 220:97–104, 2004. doi: 10.1007/s00425-004-1328-0.

T. S. Korneliussen, A. Albrechtsen, and R. Nielsen. ANGSD: analysis of next generation sequencing data. *BMC Bioinformatics*, 15(1):356, 2014.

A. Lavell, M. Smith, Y. Xu, J. E. Froehlich, C. De La Mora, and C. Benning. Proteins associated with the *Arabidopsis thaliana* plastid rhomboid-like protein RBL10. *The Plant Journal*, 108 (5):1332–1345, 2021. doi: 10.1111/tpj.15514.

P. J. Lea, L. Sodek, M. A. J. Parry, P. R. Shewry, and N. G. Halford. Asparagine in plants. *Annals of Applied Biology*, 150 (1):1–26, 2007. doi: 10.1111/j.1744-7348.2006.00104.x.

S. Lehmann, C. Gumy, E. Blatter, S. Boeffel, W. Fricke, and D. Rentsch. In planta function of compatible solute transporters of the AtProT family. *Journal of Experimental Botany*, 62(2):787–796, 2011. doi: 10.1093/jxb/erq320.

S. Léran et al. A unified nomenclature of NITRATE TRANSPORTER 1/PEPTIDE TRANSPORTER family members in plants. *Trends in Plant Science*, 19(1):5–9, 2014. doi: 10.1016/j.tplants.2013.08.008.

J. Liu, Y. H. Wu, J. J. Yang, Y. D. Liu, and F. F. Shen. Protein degradation and nitrogen remobilization during leaf senescence. *Journal of Plant Biology*, 51(1):11–19, 2008. doi: 10.1007/BF03030735.

G. Lou, P. Chen, H. Zhou, P. Li, J. Xiong, S. Wan, Y. Zheng, M. Alam, R. Liu, Y. Zhou, H. Yang, Y. Tian, J. Bai, W. Rao, X. Tan, H. Gao, Y. Li, G. Gao, Q. Zhang, X. Li, C. Liu, and Y. He. FLOURY ENDOSPERM19 encoding a class I glutamine amidotransferase affects grain quality in rice. *Molecular Breeding*, 41:36, 2021. doi: 10.1007/s11032-021-01226-z.

P. Ma, X. Zhang, B. Luo, Z. Chen, X. He, H. Zhang, B. Li, D. Liu, L. Wu, S. Gao, D. Gao, S. Zhang, and S. Gao. Transcriptomic and genome-wide association study reveal long noncoding RNAs responding to nitrogen deficiency in maize. *BMC Plant Biology*, 21:93, 2021. doi: 10.1186/s12870-021-02847-4.

C. Masclaux-Daubresse, F. Daniel-Vedele, J. Dechorgnat, F. Chardon, L. Gaufichon, and A. Suzuki. Nitrogen uptake, assimilation and remobilization in plants: challenges for sustainable and productive agriculture. *Annals of Botany*, 105(7):1141–1157, 2010a. doi: 10.1093/aob/mcq028.

C. Masclaux-Daubresse, F. Daniel-Vedele, J. Dechorgnat, F. Chardon, L. Gaufichon, and A. Suzuki. Nitrogen uptake, assimilation and remobilization in plants: challenges for sustainable and productive agriculture. *Journal of Experimental Botany*, 61(9):2285–2303, 2010b.

C. H. McAllister, M. Wolansky, and A. G. Good. The impact on nitrogen-efficient phenotypes when aspartate aminotransferase is expressed tissue-specifically in *Brassica napus*. *New Negatives in Plant Science*, 3–4:1–9, 2016. doi: 10.1016/j.neps.2016.03.001.

J. Meisner and A. Albrechtsen. Inferring population structure and admixture proportions in low-depth NGS data. *Genetics*, 210 (2):719–731, 2018. doi: 10.1534/genetics.118.301336.

Y. Miyashita and A. G. Good. NAD(H)-dependent glutamate dehydrogenase is essential for the survival of Arabidopsis thaliana during dark-induced carbon starvation. *Journal of Experimental Botany*, 59(3):667–680, 2008. doi: 10.1093/jxb/erm340.

R. H. Moll and C. W. Stuber. Comparisons of response to alternative selection procedures initiated with two populations of maize (*zea mays* L.). *Crop Science*, 11(5):706–711, 1971.

R. H. Moll, W. A. Jackson, and R. L. Mikkelsen. Recurrent selection for maize grain yield: Dry matter and nitrogen accumulation and partitioning changes. *Crop Science*, 34(4):874–881, 1994. doi: 10.2135/cropsci1994.0011183X003400040009x.

X. Mu, Q. Chen, F. Chen, L. Yuan, and G. Mi. A RNA-Seq analysis of the response of photosynthetic system to low nitrogen supply in maize leaf. *International Journal of Molecular Sciences*, 18(12):2624, 2017. doi: 10.3390/ijms18122624.

S. M. Mueller, C. D. Messina, and T. J. Vyn. Simultaneous gains in grain yield and nitrogen efficiency over 70 years of maize genetic improvement. *Scientific Reports*, 9:9095, 2019. doi: 10.1038/s41598-019-45485-5.

B. Müller, A. Fastner, J. Karmann, V. Mansch, T. Hoffmann, W. Schwab, M. Suter-Grotemeyer, D. Rentsch, E. Truernit, F. Ladwig, A. Bleckmann, T. Dresselhaus, and U. Z. Hammes. Amino acid export in developing *Arabidopsis* seeds depends on UmamiT facilitators. *Current Biology*, 25(23):3126–3131, 2015. doi: 10.1016/j.cub.2015.10.038.

B. Neuhäuser, M. Dynowski, and U. Ludewig. Channel-like NH3 flux by ammonium transporter AtAMT2. *FEBS Letters*, 583 (17):2833–2838, 2009. doi: 10.1016/j.febslet.2009.07.039.

Y. Ohkubo, M. Tanaka, R. Tabata, M. Ogawa-Ohnishi, and Y. Matsubayashi. Shoot-to-root mobile polypeptides involved in systemic regulation of nitrogen acquisition. *Nature Plants*, 3(4):17029, 2017. doi: 10.1038/nplants.2017.29.

J. O. Ojeda-Rivera, A. C. Barnes, E. A. Ainsworth, R. Angelovici, B. Basso, L. J. Brindisi, M. D. Brooks, W. Busch, G. L. Buttelmann, M. J. Castellano, J. Chen, D. E. Costich, N. De Leon, B. D. Emmett, D. Ertl, S. L. Fitzsimmons, S. A. Flint-Garcia, M. A. Gore, K. Guan, C. O. Hale, S. Herr, C. N. Hirsch, D. H. Holding, J. B. Holland, S. K. Hsu, J. Hua, M. B. Hufford, S. M. Kaeppler, E. N. Leary, Z. Y. Liu, A. A. Mahama, T. J. McCubbin, C. D. Messina, T. P. Michael, S. J. Miller, S. C. Murray, S. Okumoto, E. Oren, A. N. Park, M. A. Piñeros, N. A. Pugh, V. Raboy, R. Rellán-Álvarez, M. C. Romay, T. Rooney, R. L. Roston, R. J. H. Sawers, J. C. Schnable, A. J. Schulz, M. P. Scott, N. M. Springer, J. D. Washburn, M. A. Zambrano, J. Zhai, J. Zou, and E. S. Buckler. Designing a nitrogen-efficient cold-tolerant maize for modern agricultural systems. *The Plant Cell*, 37(7):koaf139, 2025. doi: 10.1093/plcell/koaf139.

M. S. Otegui, Y.-S. Noh, D. E. Martínez, M. G. Vila Petroff, L. A. Staehelin, R. M. Amasino, and J. J. Guiamet. Senescence-associated vacuoles with intense proteolytic activity develop in leaves of Arabidopsis and soybean. *The Plant Journal*, 41(6): 831–844, 2005. doi: 10.1111/j.1365-313X.2005.02346.x.

K. Patterson, L. A. Walters, A. M. Cooper, J. G. Olvera, M. A. Rosas, A. G. Rasmusson, and M. A. Escobar. Nitrate-regulated glutaredoxins control *Arabidopsis* primary root growth. *Plant Physiology*, 170(2):989–999, 2016. doi: 10.1104/pp.15.01776.

L. Poggio, L. M. de Sousa, N. H. Batjes, G. Heuvelink, B. Kempen, E. Ribeiro, and D. Rossiter. SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *SOIL*, 7(1):217–240, 2021. doi: 10.5194/soil-7-217-2021.

S. Potter and U. Feller. Chloroplast protein degradation pathways during leaf senescence. *Plants*, 15(17):2576, 2026.

M. S. Rasmussen, G. Garcia-Erill, T. S. Korneliussen, C. Wiuf, and A. Albrechtsen. Estimation of site frequency spectra from low-coverage sequencing data using stochastic EM reduces overfitting, runtime, and memory usage. *Genetics*, 222(4): iyac148, 2022. doi: 10.1093/genetics/iyac148.

J. A. Romero Navarro, M. Willcox, J. Burgueño, C. Romay, K. Swarts, S. Trachsel, E. Preciado, A. Terron, H. V. Delgado, V. Vidal, A. Ortega, A. E. Banda, N. O. G. Montiel, I. Ortiz-Monasterio, F. S. Vicente, A. Espinoza, G. Atlin, P. Wenzl, S. Hearne, and E. S. Buckler. A study of allelic diversity underlying flowering-time adaptation in maize landraces. *Nature Genetics*, 49(3):476–480, 2017. doi: 10.1038/ng.3784.

J. Santiago and M. Tegeder. Amino acid transporters in plant cells: A brief review. *Plant and Cell Physiology*, 61(7):1217–1228,

2020.

V. Shrestha et al. Multiomics approach reveals a role of translational machinery in shaping maize kernel amino acid composition. *Plant Physiology*, 188(1):111–133, 2022.

L. Skotte, T. S. Korneliussen, and A. Albrechtsen. Estimating individual admixture proportions from next generation sequencing data. *Genetics*, 195(3):693–702, 2013. doi: 10.1534/genetics.113.154138.

M. L. Slaten, Y. O. Chan, V. Shrestha, A. E. Lipka, and R. Angelovici. HAPPI GWAS: holistic analysis with pre- and post-integration GWAS. *Bioinformatics*, 36(17):4655–4657, 2020.

A. Soares, S. M. Ribeiro Carlton, and I. Simões. Atypical and nucellin-like aspartic proteases: emerging players in plant developmental processes and stress responses. *Journal of Experimental Botany*, 70(7):2059–2076, 2019. doi: 10.1093/jxb/erz034.

I. Teixeira, M. Santos, and M. H. Carvalho. Amino acids as multifunctional molecules in plants. *Plants*, 15(10):1583, 2026.

P. D. Thomas, D. Ebert, A. Muruganujan, T. Mushayahama, L.-P. Albou, and H. Mi. PANTHER: Making genome-scale phylogenetics accessible to all. *Protein Science*, 31(1):8–22, 2022. doi: 10.1002/pro.4218.

J. V. Torres-Rodríguez, M. N. Salazar-Vidal, R. A. Chávez Montes, J. A. Massange-Sánchez, C. S. Gillmor, and R. J. H. Sawers. Low nitrogen availability inhibits the phosphorus starvation response in maize (*zea mays* ssp. *mays* L.). *BMC Plant Biology*, 21(1):259, 2021. doi: 10.1186/s12870-021-02997-5.

R. E. Viola. The central enzymes of the aspartate family of amino acid biosynthesis. *Accounts of Chemical Research*, 34(5):339–349, 2001. doi: 10.1021/ar000057q.

Q. Wang, M. Wang, A.-A. Xia, J.-Y. Wang, Z. Wang, T. Xu, D.-T. Jia, M. Lu, W.-M. Tan, J.-H. Luo, and Y. He. Natural variation in *ZmNRT2.5* modulates husk leaf width and promotes seed protein content in maize. *Plant Biotechnology Journal*, 23(4):1039–1052, 2025. doi: 10.1111/pbi.14559.

W. Wang, M. Xu, G. Wang, and G. Galili. New insights into the metabolism of aspartate-family amino acids in plant seeds. *Plant Reproduction*, 31:203–211, 2018. doi: 10.1007/s00497-018-0322-9.

Y. Wei, S. Xiong, Z. Zhang, X. Meng, L. Wang, X. Zhang, M. Yu, H. Yu, X. Wang, and X. Ma. Localization, gene expression, and functions of glutamine synthetase isozymes in wheat grain (Triticum aestivum l.). *Frontiers in Plant Science*, 12:580405, 2021. doi: 10.3389/fpls.2021.580405.

Y. Xiong, A. L. Contento, and D. C. Bassham. AtATG18a is required for the formation of autophagosomes during nutrient stress and senescence in Arabidopsis thaliana. *The Plant Journal*, 42(4):535–546, 2005. doi: 10.1111/j.1365-313X.2005.02397.x.

A. Yobi and R. Angelovici. A high-throughput absolute-level quantification of protein-bound amino acids in seeds. *Current Protocols in Plant Biology*, 3(4):e20084, 2018. doi: 10.1002/cppb.20084.

Q. Zhang and F. Marsolais. Identification and characterization of omega-amidase as an enzyme metabolically linked to asparagine transamination in Arabidopsis. *Phytochemistry*, 99:36–43, 2014.

Y. Zhou, H. Cai, J. Xiao, X. Li, Q. Zhang, and X. Lian. Over-expression of aspartate aminotransferase genes in rice resulted in altered nitrogen metabolism and increased amino acid content in seeds. *Theoretical and Applied Genetics*, 118:1381–1390, 2009. doi: 10.1007/s00122-009-0988-3.

R. J. Zomer, J. Xu, and A. Trabucco. Version 3 of the global aridity index and potential evapotranspiration database. *Scientific Data*, 9(1):409, 2022. doi: 10.1038/s41597-022-01493-1.